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**DISCRETE EMPIRICAL
INTERPOLATION METHOD**

Master's thesis

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This Master's thesis was defended on _____ before an examination committee with the following composition:

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To everyone who supported me and kept me company throughout this journey.

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Introduction

High-fidelity mathematical models are now used throughout engineering to study systems for which physical experiments are expensive, slow, or difficult to repeat. After spatial discretization, a partial differential equation may produce a system with thousands or even millions of unknowns. A single simulation may already be costly, and the expense becomes especially important when the model must be solved repeatedly for different parameters, inputs, or design choices.

Model order reduction seeks a faithful surrogate of the full model. The aim is to preserve the input-output behaviour relevant to the application while reducing the computational cost. For a semidiscrete system written schematically as

$$\mathbf{M}\dot{x}(t; \mu) = f(x(t; \mu), t; \mu), \quad (1)$$

the state is $x(t; \mu) \in \mathbb{R}^n$, the time variable satisfies $t \in [0, T]$, and the parameter vector μ belongs to an admissible parameter set $\mathcal{D} \subseteq \mathbb{R}^{n_\mu}$. Furthermore, $\mathbf{M} \in \mathbb{R}^{n \times n}$ is a nonsingular matrix and

$$f : \mathbb{R}^n \times [0, T] \times \mathcal{D} \longrightarrow \mathbb{R}^n$$

is the possibly nonlinear right-hand side. A projection method approximates the state in a low-dimensional affine space,

$$x(t; \mu) \approx x_0 + \mathbf{V}_r \widehat{x}(t; \mu), \quad r \ll n,$$

where $\mathbf{V}_r \in \mathbb{R}^{n \times r}$ is a judiciously selected basis matrix. Proper orthogonal decomposition (POD) constructs such a space from solution snapshots and is optimal for the associated least-squares projection problem. This reduces the number of state variables, but for a general nonlinear mapping it does not by itself reduce the cost of evaluating

$$f(x_0 + \mathbf{V}_r \widehat{x}, t; \mu).$$

Forming this vector may still require work proportional to the full dimension n . A reduced system can therefore remain coupled to a full-dimensional nonlinear evaluation.

The discrete empirical interpolation method (DEIM) addresses this remaining bottleneck [5]. A low-dimensional basis is constructed for the vectors to be approximated, and

interpolation indices are selected so that the coefficients in this basis can be recovered from only a small number of vector entries. DEIM therefore combines two separate ingredients: an approximation subspace and a rule for selecting informative coordinates. Its accuracy depends both on how well the subspace represents the relevant vectors and on the stability of the selected interpolation problem.

This thesis studies both ingredients and several modifications of the DEIM workflow. The original greedy selection is first developed together with its error analysis. QDEIM replaces this sequential index selection with a column-pivoted QR factorization of the transposed basis [9]. Since in some applications sampling particular coordinates may not be feasible, we also consider selection from a prescribed admissible index set and derive the rank condition under which such a restricted interpolation problem is possible. This gives a direct mathematical model for inaccessible sensor locations, unavailable measurements, or nonlinear-vector entries that cannot be evaluated economically. Index selection is only one part of the offline construction. Conventional POD–DEIM requires snapshots of both the solution and the nonlinear term, followed by a separate compression of each snapshot matrix. The solution-based nonlinear subspace method (SNS) [7] changes this basis-construction stage. It uses the time-discrete equations to construct a nonlinear approximation space from an extended solution basis, thereby avoiding a separate nonlinear-snapshot singular value decomposition. SNS does not replace the sampling procedure: after the nonlinear subspace has been constructed, interpolation indices must still be chosen. This distinction between basis construction and coordinate selection is maintained throughout the thesis.

Although DEIM was introduced for nonlinear model reduction, its interpolation operator is not limited to nonlinear right-hand sides. The same mathematics can reconstruct a high-dimensional field from sparse coordinate values. In that setting the selected indices represent physical measurement or sensor locations. This second interpretation is useful for connecting the algebraic selection problem with practical questions such as spatial accessibility and sparse flow-field reconstruction.

The thesis is organized as follows. The first chapter develops the necessary linear algebra, SVD, column-pivoted QR factorization, and POD before presenting DEIM and its error analysis. It then treats QDEIM, restricted interpolation indices, and SNS, with particular attention to the distinct roles of approximation spaces and coordinate selection.

The second chapter examines these ideas numerically. A parameter-dependent two-dimensional function visualizes DEIM, QDEIM, and restricted interpolation points. The FitzHugh–Nagumo system provides a standard POD–DEIM evolution test problem. A nonlinear RC-ladder circuit shows how selected entries of a locally coupled vector field can be evaluated through an exact dependency stencil and compares conventional and SNS-based reduced models. Finally, a parametric cylinder-wake dataset is used to reconstruct vorticity fields from sparse spatial values on held-out and independently generated tra-

jectories. This last experiment concerns sparse state reconstruction, not a hyper-reduced Navier–Stokes time integrator.

Chapter 1

Theoretical Background

1.1 Linear Algebra Prerequisites

We first review the much needed linear algebra background for this thesis. All the definitions and results in the following sections are standard and can be easily found in the literature. A broader treatment of these standard topics can be found in [10].

Definition 1.1.1. Let $A \in \mathbb{C}^{m \times n}$. We define the following norms:

- $\|A\|_2 = \sup_{x \in \mathbb{C}^n, x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}$, the spectral norm, or matrix 2-norm, of A .
- $\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}$, the Frobenius norm of A .

From the definition of the spectral norm, it is clear that

$$\|A\|_2 = \sup_{x \in \mathbb{C}^n, \|x\|_2=1} \|Ax\|_2.$$

Definition 1.1.2. A matrix $H \in \mathbb{C}^{n \times n}$ is called Hermitian if

$$H^* = H.$$

A Hermitian matrix H is called positive semidefinite if

$$x^* H x \geq 0 \quad \text{for every } x \in \mathbb{C}^n.$$

One of the most common mathematical tools we will be using in our models is the SVD. Proof of the existence of this decomposition will be provided in the following section and it is going to be based on the following theorem.

Theorem 1.1.3 (Spectral theorem for Hermitian matrices). *Let $H \in \mathbb{C}^{n \times n}$ be Hermitian. Then there exist a unitary matrix $V \in \mathbb{C}^{n \times n}$ and a real diagonal matrix*

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$$

such that

$$H = V\Lambda V^*.$$

Equivalently, $\mathbb{C}^n$ possesses an orthonormal basis consisting of eigenvectors of H . Moreover, H is positive semidefinite if and only if

$$\lambda_i \geq 0, \quad i = 1, \dots, n.$$

Lemma 1.1.4. *Let $A \in \mathbb{R}^{m \times n}$ be a matrix with full column rank. Then $A^T A$ is invertible, and the matrix*

$$P = A(A^T A)^{-1} A^T \in \mathbb{R}^{m \times m}$$

is the orthogonal projector onto the column space of A .

Proof. By the Rank-Nullity Theorem, it is enough to show that for every $x \in \mathbb{R}^n$, having $A^T A x = 0$ implies $x = 0$. Let's assume $A^T A x = 0$ for some $x \in \mathbb{R}^n$. Multiplying by x^T on the left, we get

$$0 = x^T A^T A x = (Ax)^T (Ax) = \|Ax\|_2^2,$$

which implies $Ax = 0$. Since A has full column rank, this means $x = 0$. Therefore, $A^T A$ is invertible. Hence, P is well-defined. We now show that P is a projector. We have

$$P^2 = A(A^T A)^{-1} A^T A(A^T A)^{-1} A^T = A(A^T A)^{-1} A^T = P.$$

Therefore, P is a projector. It remains to identify its range. For any $x \in \mathbb{R}^m$, we have

$$Px = A(A^T A)^{-1} A^T x \in \text{Ran}(A),$$

so

$$\text{Ran}(P) \subseteq \text{Ran}(A).$$

Conversely, let $y \in \text{Ran}(A)$. Then there exists $z \in \mathbb{R}^n$ such that $y = Az$. Hence

$$Py = A(A^T A)^{-1} A^T Az = Az = y.$$

This implies that $y \in \text{Ran}(P)$. Therefore,

$$\text{Ran}(P) = \text{Ran}(A).$$

Finally, we show that the projector is orthogonal. Since $A^T A$ is symmetric, its inverse is also symmetric. Therefore,

$$P^T = (A(A^T A)^{-1} A^T)^T = A((A^T A)^{-1})^T A^T = A(A^T A)^{-1} A^T = P.$$

Thus P is symmetric. Since P is a symmetric projector onto $\text{Ran}(A)$, it is the orthogonal projector onto $\text{Ran}(A)$. $\square$

The previous lemma is a standard linear algebra result for the least squares problem

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|_2.$$

It formalises finding the vector in the column space of A closest to b . The solution satisfies the normal equations

$$A^T A x = A^T b,$$

and the lemma gives $x = (A^T A)^{-1} A^T b$, so the closest vector is $Ax = A(A^T A)^{-1} A^T b$.

Definition 1.1.5 (Householder Reflector). *Let $v \in \mathbb{C}^n$. The Householder reflector generated by v is the matrix $H_v \in \mathbb{C}^{n \times n}$ defined by*

$$H_v = \begin{cases} \mathbf{I} - 2 \frac{vv^*}{\|v\|_2^2}, & v \neq 0, \\ \mathbf{I}, & v = 0. \end{cases}$$

The matrix H_v is Hermitian, since

$$(H_v)^* = \left(\mathbf{I} - 2 \frac{vv^*}{\|v\|_2^2} \right)^* = \mathbf{I} - 2 \frac{vv^*}{\|v\|_2^2} = H_v.$$

Moreover, H_v is unitary. Indeed,

$$\begin{aligned} (H_v)^* H_v &= \left(\mathbf{I} - 2 \frac{vv^*}{\|v\|_2^2} \right) \left(\mathbf{I} - 2 \frac{vv^*}{\|v\|_2^2} \right) \\ &= \mathbf{I} - 4 \frac{vv^*}{\|v\|_2^2} + 4 \frac{vv^* vv^*}{\|v\|_2^4}. \end{aligned}$$

Since

$$vv^* vv^* = v(v^* v)v^* = \|v\|_2^2 vv^*,$$

we obtain

$$\begin{aligned} (H_v)^* H_v &= \mathbf{I} - 4 \frac{vv^*}{\|v\|_2^2} + 4 \frac{\|v\|_2^2 vv^*}{\|v\|_2^4} \\ &= \mathbf{I} \end{aligned}$$

which proves the claim.

Remark 1.1.6. If $x, y \in \mathbb{R}^n$ satisfy $\|x\|_2 = \|y\|_2$, then there exists a Householder reflector mapping x to y . In particular, if $x \neq y$, one may take $v = x - y$, so that

$$H_v x = y.$$

In the trivial case $x = y$, one takes the identity matrix $\mathbf{I}$.

Remark 1.1.7. Householder reflector is used in QR factorization to transform a given vector into a multiple of the first coordinate vector. Let $x \in \mathbb{R}^\ell$ be nonzero. We want a reflector H_v such that

$$H_v x = \xi e_1,$$

where

$$\xi = \pm \|x\|_2.$$

A common numerically stable choice is

$$\xi = -\text{sign}(x_1)\|x\|_2,$$

with the convention $\text{sign}(0) = 1$. Then we set

$$v = x - \xi e_1.$$

The Householder reflector generated by v maps x to ξe_1 , and therefore annihilates all components of x except the first one.

In QR factorization, this construction is applied to one active column at a time. At the k -th step, the reflector acts only on the active part of the column, namely on rows $k, \dots, m$. Therefore it is embedded into a larger orthogonal matrix of the form

$$Q_k = \begin{bmatrix} I_{k-1} & 0 \\ 0 & H_k \end{bmatrix}.$$

Thus Q_k leaves the first $k-1$ rows unchanged and eliminates the entries below the diagonal in the k -th column.

Remark 1.1.8. To end on a rigorous note, we will explain why we choose

$$\xi = -\text{sign}(x_1)\|x\|_2$$

over $\xi = \text{sign}(x_1)\|x\|_2$ in the Householder reflector.

For either ξ we have

$$\|v\|_2^2 = \|x - \xi e_1\|_2^2 = \|x\|_2^2 + |\xi|^2 - 2 \text{Re}(\xi^* x_1).$$

If we choose $\xi = -\text{sign}(x_1)\|x\|_2$, then

$$\|v\|_2^2 = \|x\|_2^2 + \|x\|_2^2 + 2\|x\|_2|x_1| = 2\|x\|_2(\|x\|_2 + |x_1|)$$

and the choice $\xi = \text{sign}(x_1)\|x\|_2$ yields

$$\|v\|_2^2 = \|x\|_2^2 + \|x\|_2^2 - 2\|x\|_2|x_1| = 2\|x\|_2(\|x\|_2 - |x_1|),$$

which can be very small if x is nearly parallel to e_1 . In this case, since $\|x\|_2 \approx |x_1|$, the first component $v_1 = x_1 - \xi$ is obtained by subtracting two nearly equal quantities, leading to subtractive cancellation in floating-point arithmetic.

1.2 Singular value decomposition (SVD)

A substantial portion of model reduction techniques relies on low-rank approximations. In the realm of matrices, a central role is played by the *Singular Value Decomposition* (SVD). A detailed treatment of its basic properties and low-rank consequences is given in [10].

Theorem 1.2.1. *Let $A \in \mathbb{C}^{m \times n}$ be a matrix of rank r . Then there exist unitary matrices $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ such that*

$$A = U\Sigma V^*, \tag{1.1}$$

where $\Sigma = \text{diag}(\sigma_i) \in \mathbb{R}^{m \times n}$ is a rectangular diagonal, with diagonal entries

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0, \quad \sigma_{r+1} = \cdots = \sigma_{\min\{m,n\}} = 0.$$

Proof. Let $B = A^*A \in \mathbb{C}^{n \times n}$. The matrix B is Hermitian, since

$$B^* = (A^*A)^* = A^*A = B,$$

and it is positive semidefinite because

$$x^*Bx = x^*A^*Ax = \|Ax\|_2^2 \geq 0 \quad \text{for every } x \in \mathbb{C}^n.$$

By the spectral theorem for Hermitian matrices, Theorem 1.1.3, there exist a unitary matrix

$$V = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$$

and real eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$ such that

$$B = V \text{diag}(\lambda_1, \dots, \lambda_n) V^*.$$

The positive semidefiniteness of B implies that $\lambda_i \geq 0$ for every i .

We first determine how many of these eigenvalues are positive. For every $x \in \mathbb{C}^n$,

$$A^*Ax = 0 \iff x^*A^*Ax = \|Ax\|_2^2 = 0 \iff Ax = 0.$$

Hence $\text{Ker}(A^*A) = \text{Ker}(A)$. The rank-nullity theorem therefore gives

$$\text{rank}(A^*A) = \text{rank}(A) = r.$$

Consequently, precisely the first r eigenvalues are positive. Define

$$\sigma_i = \sqrt{\lambda_i}, \quad i = 1, \dots, r.$$

Then $\sigma_1 \geq \dots \geq \sigma_r > 0$.

For each $i = 1, \dots, r$, define

$$u_i = \frac{Av_i}{\sigma_i}.$$

These vectors are orthonormal. Indeed, for $1 \leq i, j \leq r$,

$$u_i^*u_j = \frac{1}{\sigma_i\sigma_j}v_i^*A^*Av_j = \frac{\lambda_j}{\sigma_i\sigma_j}v_i^*v_j = \delta_{ij}.$$

Complete $\{u_1, \dots, u_r\}$ to an orthonormal basis $\{u_1, \dots, u_m\}$ of $\mathbb{C}^m$, and let

$$U = \begin{bmatrix} u_1 & \dots & u_m \end{bmatrix}.$$

Thus, U is unitary. Let $\Sigma \in \mathbb{R}^{m \times n}$ be the rectangular diagonal matrix whose first r diagonal entries are $\sigma_1, \dots, \sigma_r$ and whose remaining entries are zero.

It remains to verify the claimed factorization. If $i \leq r$, then

$$U\Sigma V^*v_i = \sigma_i u_i = Av_i.$$

If $i > r$, then $\lambda_i = 0$ and

$$\|Av_i\|_2^2 = v_i^*A^*Av_i = \lambda_i = 0,$$

so both $U\Sigma V^*v_i$ and Av_i are zero. Therefore, the two linear maps $U\Sigma V^*$ and A agree on the orthonormal basis $\{v_1, \dots, v_n\}$ of $\mathbb{C}^n$, and hence

$$A = U\Sigma V^*.$$

The same construction also covers $r = 0$: in that case $A = 0$, one may choose any unitary matrices U and V , and take $\Sigma = 0$. $\square$

Remark 1.2.2.

- The scalars σ_i are called the singular values of A .
- The columns of U are called the left singular vectors of A , and the first r columns of U form an orthonormal basis for the column space of A .
- The columns of V are called the right singular vectors of A , and the last $n-r$ columns of V form an orthonormal basis for $\text{Ker}(A)$, the null space of A .

Rather than solving a maximization problem, SVD gives a practical way to compute the spectral norm of a matrix defined in the preceding section.

Lemma 1.2.3. *Let $A \in \mathbb{C}^{m \times n}$ and $A = U\Sigma V^*$ be the singular value decomposition of A . Then*

$$\|A\|_2 = \sigma_{\max}(A),$$

where $\sigma_{\max}(A)$ is the largest singular value of A .

Proof. Since U and V are unitary, they preserve the Euclidean norm. Hence, for every $x \in \mathbb{C}^n$ with $\|x\|_2 = 1$, if we set $y = V^*x$, then $\|y\|_2 = 1$ and

$$\|Ax\|_2 = \|U\Sigma V^*x\|_2 = \|\Sigma y\|_2.$$

Therefore,

$$\|A\|_2 = \sup_{\|y\|_2=1} \|\Sigma y\|_2.$$

Since Σ is diagonal, with the singular values of A on the diagonal, we have

$$\|\Sigma y\|_2^2 = \sum_i \sigma_i^2 |y_i|^2 \leq \sigma_{\max}(A)^2 \sum_i |y_i|^2 = \sigma_{\max}(A)^2.$$

Thus $\|A\|_2 \leq \sigma_{\max}(A)$. Since the singular values are ordered, equality is attained by choosing $y = e_1$. Therefore,

$$\|A\|_2 = \sigma_{\max}(A).$$

□

We will refer to the above factorization as the **full** SVD. In practice, it is often convenient to work with the **reduced** SVD (also called the compact SVD), obtained by removing the zero singular values. If a matrix A has rank r and its singular value decomposition is given by

$$A = U\Sigma V^*,$$

then the reduced SVD is given by

$$A = U_r \Sigma_r V_r^*, \tag{1.2}$$

where U_r and V_r consist of the first r columns of U and V , respectively, and $\Sigma_r \in \mathbb{R}^{r \times r}$ is the diagonal matrix containing all nonzero singular values of A .

Before turning to low-rank approximation, we record two structural facts. The singular values are intrinsic to the matrix, whereas individual singular vectors need not be unique. Moreover, multiplying by a unitary matrix preserves the corresponding singular values and one family of singular subspaces.

Lemma 1.2.4. *Let $A \in \mathbb{C}^{m \times n}$. The singular values of A are uniquely determined by A , but the left and right singular vectors in a singular value decomposition are not unique in general.*

Proof. Let $A \in \mathbb{C}^{m \times n}$ and let

$$A = U\Sigma V^*$$

be a singular value decomposition of A . Let u_i and v_i be left and right singular vectors corresponding to a nonzero singular value σ_i . Then

$$Av_i = \sigma_i u_i.$$

Let $\alpha \in \mathbb{C}$ satisfy $|\alpha| = 1$. Then

$$A(\alpha v_i) = \alpha Av_i = \alpha \sigma_i u_i = \sigma_i (\alpha u_i).$$

Hence αu_i is also a left singular vector corresponding to the same singular value σ_i . Since α can be chosen with $\alpha \neq 1$, the left singular vector is not unique. The same argument applies to the right singular vectors, since the right singular vectors of A are the left singular vectors of A^* . To prove uniqueness of the singular values, consider A^*A . We have

$$A^*A = (U\Sigma V^*)^* U\Sigma V^* = V\Sigma^* \Sigma V^*.$$

Since $\Sigma^* \Sigma$ is diagonal with the squares of the singular values of A on the diagonal, the singular values of A are uniquely determined by the eigenvalues of A^*A , which are uniquely determined by A . $\square$

Lemma 1.2.5. *Let $A \in \mathbb{C}^{m \times n}$. If $Q \in \mathbb{C}^{n \times n}$ is unitary, then A and AQ have the same singular values, and their left singular vectors can be chosen to be the same. Equivalently, they have the same left singular subspaces. Similarly, if $P \in \mathbb{C}^{m \times m}$ is unitary, then A and PA have the same singular values and the same right singular subspaces.*

Proof. Let

$$A = U\Sigma V^*$$

be a singular value decomposition of A . Then

$$AQ = U\Sigma V^* Q = U\Sigma (Q^* V)^*.$$

Since Q^*V is unitary, the above is a singular value decomposition of AQ with the same left singular vectors and the same singular values as the SVD of A . The same argument gives

$$PA = PU\Sigma V^*,$$

where PU is unitary, so PA has the same right singular vectors and the same singular values. $\square$

More generally, one may keep only the first $k < r$ singular values and singular vectors. The following result states that this truncation gives the best possible approximation of rank at most k ; see also [10].

Theorem 1.2.6 (Eckart-Young-Mirsky-Schmidt). *Let $A \in \mathbb{C}^{m \times n}$ be a matrix of rank r with singular value decomposition $A = U\Sigma V^*$ and let $k \in \mathbb{N}$, $k < r$. If we define*

$$A_k = U_k \Sigma_k V_k^*,$$

where U_k and V_k consist of the first k columns of U and V , respectively, and $\Sigma_k \in \mathbb{R}^{k \times k}$ is the leading $k \times k$ block of Σ , then

$$\begin{aligned} \|A - A_k\|_2 &= \min_{\text{rank}(B) \leq k} \|A - B\|_2 = \sigma_{k+1}. \\ \|A - A_k\|_F &= \min_{\text{rank}(B) \leq k} \|A - B\|_F = \sqrt{\sigma_{k+1}^2 + \cdots + \sigma_r^2}. \end{aligned}$$

The above result can be illustrated by a simple image compression experiment. A grayscale image of size $m \times n$ may be regarded as a matrix $A \in \mathbb{R}^{m \times n}$, where the entry A_{ij} represents the intensity of the pixel in the i -th row and j -th column. Instead of storing all mn entries of A , we approximate it by

$$A_k = U_k \Sigma_k V_k^*,$$

which has rank at most k . This representation is low-rank because it is built from only the first k singular directions; storing U_k , Σ_k and V_k requires approximately $k(m + n + 1)$ numbers. Thus, when $k(m + n + 1) \ll mn$ and the neglected singular values are small, the image can be stored using much less data while retaining most of its visible structure. In the example below, the resized grayscale image has size 900×600 , so the full matrix contains 540000 intensity values. A rank- k SVD approximation stores $k(900 + 600 + 1)$ values; for instance, $k = 20$ requires only 30020 values, about 5.6% of the original storage count. The singular value decay in Figure 1.1 explains why such a low-rank approximation is visually reasonable for this image. Using only the first k singular values and singular vectors gives the reconstructions shown in Figure 1.2.

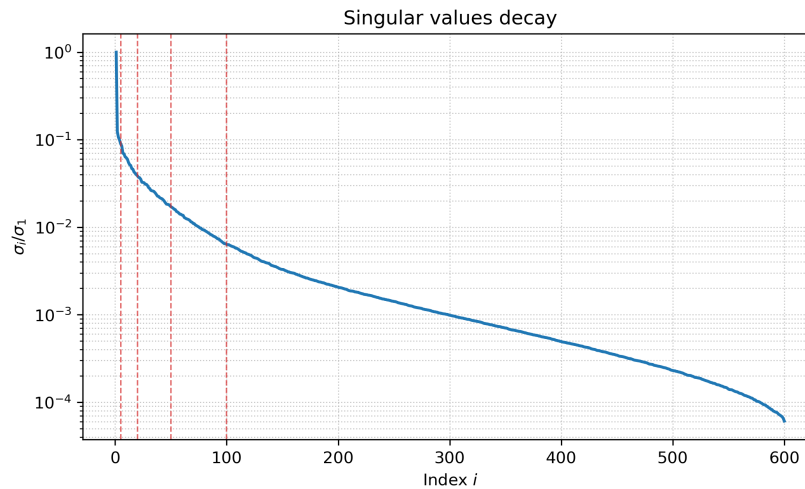


Figure 1.1: Decay of the singular values of the grayscale image, normalized by the largest singular value. The dashed vertical lines mark the values of k used in the reconstructions.

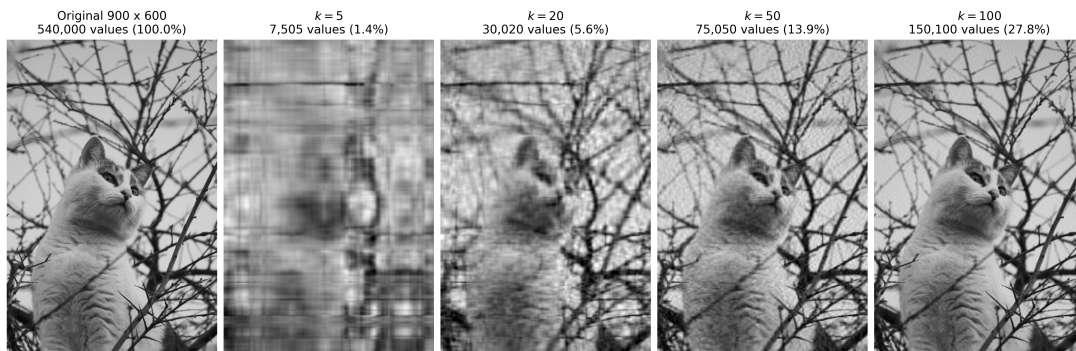


Figure 1.2: Image reconstruction using truncated SVD. The original image is shown on the left, followed by reconstructions using $k = 5, 20, 50, 100$ singular values. The titles indicate the number of stored values and the percentage relative to the original grayscale matrix.

1.3 Column-Pivoted QR Decomposition

We now turn from the optimal low-rank approximations provided by the SVD to a cheaper factorization that is often used in practice to reveal numerical rank and select representative columns. The construction uses the Householder reflections introduced in the linear alge-

bra prerequisites, together with column pivoting. The role of QR factorization and column pivoting in least-squares problems and numerical rank determination is discussed in [10].

Theorem 1.3.1 (Businger–Golub Column-Pivoted QR Decomposition). *Let $A \in \mathbb{R}^{m \times n}$. The Businger–Golub column-pivoted QR algorithm [4] produces a permutation matrix $P \in \mathbb{R}^{n \times n}$ and an orthogonal matrix $Q \in \mathbb{R}^{m \times m}$ such that*

$$AP = QR,$$

where $R \in \mathbb{R}^{m \times n}$ has the block form

$$R = \begin{bmatrix} R_{11} & R_{12} \\ 0 & 0 \end{bmatrix},$$

Writing $r = \text{rank}(A)$, the blocks satisfy $R_{11} \in \mathbb{R}^{r \times r}$ and $R_{12} \in \mathbb{R}^{r \times (n-r)}$, while the two lower zero blocks have dimensions $(m-r) \times r$ and $(m-r) \times (n-r)$, respectively. Blocks with zero rows or columns are omitted. Thus, for full column rank, $R = [R_{11}^T \ 0^T]^T$, whereas for full row rank, $R = [R_{11} \ R_{12}]$. The matrix R_{11} is upper triangular and nonsingular. Moreover, the diagonal entries of R_{11} are nonincreasing in absolute value and positive.

Proof. We describe the construction. Set

$$A^{(1)} = A.$$

Let $p = \min\{m, n\}$. Each step creates one new diagonal entry of R , and an $m \times n$ matrix has p diagonal entries. Therefore, the algorithm can perform at most p steps. If $A = 0$, the result is immediate, so suppose that $A \neq 0$.

At the first step, the Businger–Golub pivoting rule selects a column of $A^{(1)}$ with maximal Euclidean norm and interchanges it with the first column. This column interchange is represented by a permutation matrix $P_1 \in \mathbb{R}^{n \times n}$. A Householder reflector is then applied from the left in order to annihilate all entries below the first entry of the first column. Thus there exists an orthogonal matrix Q_1 such that

$$Q_1^T A^{(1)} P_1 = \begin{bmatrix} r_{11} & * & \cdots & * \\ 0 & & & \\ \vdots & & A_{\text{act}}^{(2)} & \\ 0 & & & \end{bmatrix}.$$

The same idea is repeated on the trailing active submatrix. Suppose that the first $k-1$ steps have already been performed. At step k , the first $k-1$ columns are already fixed, and

pivoting is performed only among the remaining columns. Therefore the column permutation at step k has the block form

$$P_k = \begin{bmatrix} I_{k-1} & 0 \\ 0 & \widehat{P}_k \end{bmatrix},$$

where $\widehat{P}_k$ is a permutation matrix acting on the remaining columns.

If the largest active column norm is zero, the algorithm terminates. Otherwise, after this column interchange, a Householder reflector is applied to the active part of the k -th column, that is, to its entries in rows $k, \dots, m$. If this active vector is denoted by x_k , we choose a Householder reflector H_k such that

$$H_k x_k = \xi_k e_1, \quad |\xi_k| = \|x_k\|_2.$$

Embedded into the full space, this reflector has the form

$$Q_k = \begin{bmatrix} I_{k-1} & 0 \\ 0 & H_k \end{bmatrix}.$$

Thus Q_k leaves the first $k-1$ rows unchanged and annihilates the entries below the diagonal in the k -th column.

Let $s \leq p$ be the number of steps performed before termination; if the algorithm does not terminate early, then $s = p$. After these steps, we obtain

$$Q_s^T \cdots Q_1^T A P_1 \cdots P_s = R,$$

where R has zeros below its first s diagonal positions. If $s < p$, the remaining active submatrix is zero. Thus R is upper trapezoidal, with its precise block structure determined below. Defining

$$P = P_1 \cdots P_s$$

and

$$Q = Q_1 \cdots Q_s,$$

we get

$$Q^T A P = R,$$

or equivalently

$$A P = Q R.$$

Since each Q_k is orthogonal, their product Q is orthogonal. Since each P_k is a permutation matrix, their product P is a permutation matrix.

We now relate the construction to the real and numerical rank of A . At the k -th step, the active part of each remaining column represents the component which has not yet been

eliminated by the previously selected pivot columns. The Businger–Golub pivoting rule chooses a remaining column whose active part has maximal Euclidean norm.

If this maximal active norm is nonzero, then the selected column has a nonzero component outside the subspace generated by the previous pivot columns. Hence it contributes a new linearly independent direction, and the corresponding diagonal entry satisfies

$$|r_{kk}| > 0.$$

If the maximal active norm is zero, then all remaining active column parts are zero. In that case, all remaining columns lie in the span of the previously selected pivot columns, and no new independent direction remains.

Consequently, the number of nonzero diagonal entries produced by the algorithm is exactly $\text{rank}(A)$. Hence R has the block form

$$R = \begin{bmatrix} R_{11} & R_{12} \\ 0 & 0 \end{bmatrix},$$

where R_{11} is an upper triangular nonsingular matrix of order $\text{rank}(A)$.

Finally, we explain the monotonicity of the diagonal entries. At step k , the pivoting rule chooses the remaining column with largest active Euclidean norm. After the Householder reflector is applied, this norm becomes $|r_{kk}|$. For the remaining columns, the next active part is obtained by removing the newly created leading component, and therefore their active norms cannot increase. Hence the next pivot satisfies

$$|r_{k+1,k+1}| \leq |r_{kk}|.$$

Thus the diagonal entries of R_{11} are nonincreasing in absolute value.

It remains only to normalize their signs. Let $r = \text{rank}(A)$ and define

$$D = \text{diag}(d_1, \dots, d_m), \quad d_i = \begin{cases} \text{sign}(r_{ii}), & 1 \leq i \leq r, \\ 1, & r < i \leq m. \end{cases}$$

Since $D^2 = I_m$, we have

$$AP = QR = (QD)(DR).$$

The matrix QD is orthogonal, while DR has the same block structure and the same absolute diagonal values as R . Its first r diagonal entries are $|r_{11}|, \dots, |r_{rr}|$ and are therefore positive. Relabelling QD and DR as Q and R completes the proof. $\square$

In numerical applications, the trailing block need not be exactly zero in order for the factorization to be useful. Suppose that, for some k , we partition

$$Q = \begin{bmatrix} Q_1 & Q_2 \end{bmatrix}, \quad R = \begin{bmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{bmatrix},$$

where Q_1 contains the first k columns of Q . Then

$$AP = Q_1 \begin{bmatrix} R_{11} & R_{12} \end{bmatrix} + Q_2 \begin{bmatrix} 0 & R_{22} \end{bmatrix}.$$

Consequently, if R_{22} is small in norm, then the columns of AP are well approximated by the subspace spanned by the columns of Q_1 . Since P is only a permutation matrix, A and AP have the same column space, and therefore Q_1 also provides an approximate orthonormal basis for the range of A . More precisely,

$$AP - Q_1 Q_1^T AP = Q_2 \begin{bmatrix} 0 & R_{22} \end{bmatrix},$$

so for any unitarily invariant norm, in particular the spectral and Frobenius norms, the projection error is equal to $\|R_{22}\|$. This is the practical rank-revealing interpretation of column-pivoted QR: a small trailing block indicates that the matrix is well approximated by a matrix of rank at most k . Unlike the truncated SVD, this approximation is not generally optimal, but it is often much cheaper to compute and is very useful for selecting representative columns.

Figure 1.3 repeats the image reconstruction example using the first k columns of Q , for the same values of k as in the SVD example.

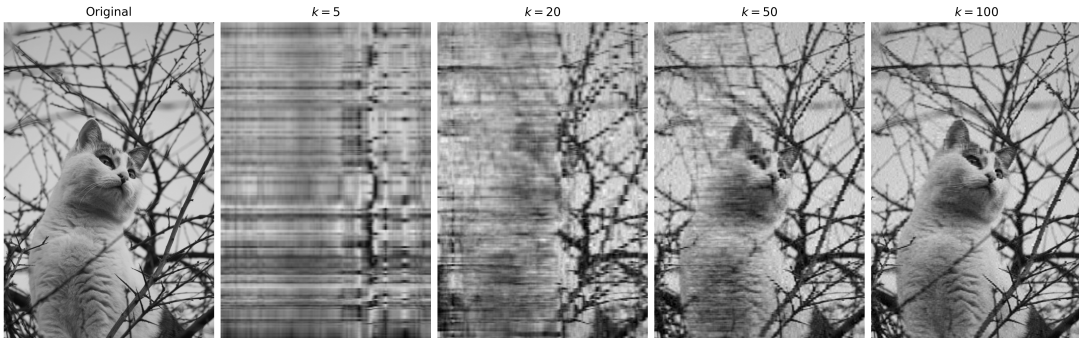


Figure 1.3: Image reconstruction using the first k columns of Q from the column-pivoted QR factorization. The original image is shown on the left, followed by reconstructions for $k = 5, 20, 50, 100$.

While there are examples where column-pivoted QR does not provide an approximation as accurate as the truncated SVD, in practice it often performs well and is widely used in applications. Figure 1.4 compares the relative Frobenius reconstruction error obtained from the truncated SVD with the projection error obtained from the first k columns of Q in the column-pivoted QR factorization, using the same image matrix as in the SVD reconstruction example. The SVD curve is lower, as expected from the Eckart–Young–Mirsky–Schmidt theorem, but the QR curve shows that the pivoted QR basis still captures much of the dominant column space.

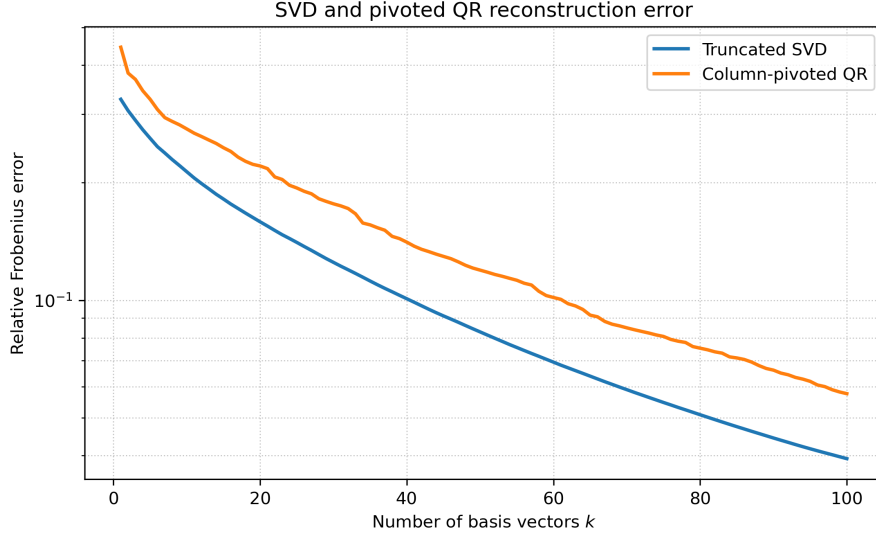


Figure 1.4: Relative Frobenius reconstruction error for the truncated SVD and the column-pivoted QR basis, for $k = 1, \dots, 100$.

1.4 Proper orthogonal decomposition (POD)

Suppose we are given a set of data snapshots $\{\mathbf{y}^1, \dots, \mathbf{y}^{n_s}\} \subset \mathbb{R}^n$, where n_s denotes the number of snapshots. In the context of dynamical systems, snapshots typically represent sampled solution states at different time instances or parameter values. Let $r \in \mathbb{N}$ be the dimension of

$$\mathcal{Y} = \text{span}\{\mathbf{y}^1, \dots, \mathbf{y}^{n_s}\}.$$

For a fixed $k \in \mathbb{N}$ with $k < r$, we consider the following minimization problem:

$$\min_{Q^T Q = \mathbb{I}_k} \sum_{i=1}^{n_s} \|\mathbf{y}^i - QQ^T \mathbf{y}^i\|_2^2. \quad (1.3)$$

Since QQ^T is the orthogonal projection onto $\text{span}(Q)$, the minimization problem is equivalent to finding the best k -dimensional subspace that approximates $\mathcal{Y}$ in the least-squares sense. A basis $\{\phi_i\}_{i=1}^k$ corresponding to the solution of the above problem is called a **POD** basis. A standard application of the singular value decomposition is that the first k left singular vectors of the snapshot matrix

$$\mathbf{Y} := [\mathbf{y}^1, \dots, \mathbf{y}^{n_s}]$$

form a POD basis. The order in which the snapshots are stored does not affect the POD minimization problem. Indeed, if $\Pi \in \mathbb{R}^{n_s \times n_s}$ is a permutation matrix, then

$$(\mathbf{Y}\Pi)(\mathbf{Y}\Pi)^T = \mathbf{Y}\mathbf{Y}^T.$$

Hence, the singular values and the minimum POD error are unchanged. If $\sigma_k > \sigma_{k+1}$, the optimal k -dimensional POD subspace is unique, so its projector is also unchanged. If $\sigma_k = \sigma_{k+1}$, the minimizer need not be unique, and an SVD routine may return a different optimal subspace within the repeated singular subspace. For example, if $\mathbf{Y} = \mathbf{I}_2$ and $k = 1$, every one-dimensional subspace is optimal. Indeed, let

$$\mathbf{Y} = \mathbf{V}_r \Sigma_r \mathbf{W}_r^T$$

be the reduced SVD of $\mathbf{Y}$. Then the optimal solution of (1.3) is given by

$$\mathbf{Q} = [\mathbf{v}_1, \dots, \mathbf{v}_k],$$

where $\mathbf{v}_i$ is the i -th column of $\mathbf{V}_r$, and the minimum value is given by

$$\min_{\mathbf{Q}^T \mathbf{Q} = \mathbf{I}_k} \sum_{i=1}^{n_s} \|\mathbf{y}^i - \mathbf{Q}\mathbf{Q}^T \mathbf{y}^i\|_2^2 = \sum_{i=k+1}^r \sigma_i^2.$$

1.5 Discrete Empirical Interpolation Method - DEIM

The discrete empirical interpolation method is a finite-dimensional variant of the empirical interpolation method (EIM), introduced by Barrault et al. for efficient reduced-basis discretizations of parametrized partial differential equations [1]. The purpose of DEIM is to reduce the cost of evaluating nonlinear vector-valued functions. In projection-based model reduction, the reduced state may have small dimension, but the evaluation of the nonlinear term may still require operations in the original high-dimensional space. Therefore, an additional approximation of the nonlinear term is needed.

1.5.1 DEIM Approximation of Nonlinear Scalar Functions

Suppose we are given a feasible parameter set $\mathcal{P}$ and a family of parameter-dependent functions

$$\{f(\cdot; p) : \mathcal{D} \rightarrow \mathbb{R} \mid p \in \mathcal{P}\},$$

where $\mathcal{D} \subseteq \mathbb{R}^k$ is the domain of the functions. For each parameter value $p \in \mathcal{P}$, the object $f(\cdot; p)$ is a scalar function defined on $\mathcal{D}$. In order to work in a finite-dimensional setting, we assume that $\mathcal{D}$ is discretized by a finite grid of n points,

$$x_1, x_2, \dots, x_n \in \mathcal{D}.$$

Then the function $f(\cdot; p)$ can be represented by the vector of its values at the grid points,

$$f(p) := \begin{bmatrix} f(x_1; p) \\ f(x_2; p) \\ \vdots \\ f(x_n; p) \end{bmatrix} \in \mathbb{R}^n.$$

In this way, after the discretization has been fixed, we identify the function $f(\cdot; p)$ with its discrete representation $f(p) \in \mathbb{R}^n$. If one wants to evaluate the function at a point which does not belong to the grid, an additional interpolation procedure may be used. However, in the DEIM setting considered here, the approximation is constructed for the discrete vector representation of the function. The mapping we are interested in is therefore

$$p \mapsto f(p) \in \mathbb{R}^n,$$

that is, we want to capture the behavior of the whole family of functions as the parameter p varies over the set $\mathcal{P}$. The idea is to approximate these high-dimensional vectors by projecting them onto an m -dimensional subspace, where $m \ll n$. This subspace is chosen so that it approximates the space generated by the nonlinear function values. Suppose we are given a basis

$$\{u_1, \dots, u_m\} \subset \mathbb{R}^n$$

for this approximation subspace. We seek an approximation of the form

$$f(p) \approx \mathbf{U}_m \mathbf{c}(p), \tag{1.4}$$

where

$$\mathbf{U}_m = \begin{bmatrix} u_1 & u_2 & \cdots & u_m \end{bmatrix} \in \mathbb{R}^{n \times m},$$

and $\mathbf{c}(p) \in \mathbb{R}^m$ is the vector of coefficients. Since this system has n equations and only m unknowns, it is overdetermined, so we select m rows to solve for $\mathbf{c}(p)$. For this purpose, define the selection matrix $P \in \mathbb{R}^{n \times m}$ by

$$P = \begin{bmatrix} e_{\varphi_1} & e_{\varphi_2} & \cdots & e_{\varphi_m} \end{bmatrix},$$

where $e_{\varphi_i} \in \mathbb{R}^n$ is the φ_i -th standard basis vector. Selecting m rows of the system (1.4) is equivalent to multiplying it from the left by P^T , which gives

$$P^T f(p) = P^T \mathbf{U}_m \mathbf{c}(p).$$

If $P^T \mathbf{U}_m$ is nonsingular, we can solve for $\mathbf{c}(p)$ and obtain

$$\mathbf{c}(p) = (P^T \mathbf{U}_m)^{-1} P^T f(p)$$

and the approximation in (1.4) becomes

$$f(p) \approx \mathbf{U}_m(P^T \mathbf{U}_m)^{-1} P^T f(p).$$

This is not yet a complete algorithm, since exactly two choices remain to be specified: how to choose the basis $\{u_1, \dots, u_m\}$ and how to choose the selection matrix P . Once these are fixed, the algorithm is deterministic. We use the POD-DEIM method to determine both the basis and the selection matrix. Suppose we are given snapshots $\{f(\cdot; p_i) : i = 1, 2, \dots, n_{train}\}$ where $p_i \in \mathcal{P}_{train}$. Define the snapshot matrix F by

$$F = \begin{bmatrix} f(\cdot; p_1) & f(\cdot; p_2) & \dots & f(\cdot; p_{n_{train}}) \end{bmatrix} \in \mathbb{R}^{n \times n_{train}}.$$

For a chosen $m \in \mathbb{N}$ with $m \ll n_{train}$, compute the SVD $F = U \Sigma V^T$ and obtain the reduced basis $\mathbf{U}_m$ by taking the first m columns of U , as motivated by POD. Before giving a precise algorithm for selecting the indices $\varphi_1, \varphi_2, \dots, \varphi_m$, let us first explain how one might arrive at such an algorithm intuitively. It is natural to expect an iterative procedure: start with one index, then add a new one where the current approximation is lacking. Since our goal is to find coefficients for which $f(p)$ is well approximated by the columns of $\mathbf{U}_m$, it is also useful to identify where each $\mathbf{u}_i$ is no longer well represented by the previously chosen indices. To formalize this, suppose that at iteration ℓ we have selected $\ell - 1$ indices and have the following objects

- $\mathbf{U}_{\ell-1} = \begin{bmatrix} u_1 & u_2 & \dots & u_{\ell-1} \end{bmatrix} \in \mathbb{R}^{n \times (\ell-1)}$
- $P_{\ell-1} = \begin{bmatrix} e_{\varphi_1} & e_{\varphi_2} & \dots & e_{\varphi_{\ell-1}} \end{bmatrix} \in \mathbb{R}^{n \times (\ell-1)}$

We then consider the next basis vector u_ℓ and examine how well the current indices approximate u_ℓ as a whole vector. First, we restrict the new basis vector to the current indices, which gives $P_{\ell-1}^T u_\ell$, and represent it as a linear combination of the current basis vectors restricted to the same indices by solving $P_{\ell-1}^T \mathbf{U}_{\ell-1} \mathbf{c} = P_{\ell-1}^T u_\ell$ for $\mathbf{c}$. Thus, the vector $\mathbf{c}$ gives the coordinates of the new vector u_ℓ in the previous basis, but only at the current indices. To measure how well these coordinates approximate the full vector, we compute the residual

$$r_{\ell-1} = u_\ell - \mathbf{U}_{\ell-1} \mathbf{c}$$

We then select the index φ_ℓ where this residual is largest, that is, where a new index is needed the most:

$$\varphi_\ell = \operatorname{argmax}_{1 \leq i \leq n} |r_{\ell-1}(i)|.$$

Finally, we update

$$P_\ell = \begin{bmatrix} P_{\ell-1} & e_{\varphi_\ell} \end{bmatrix}, \quad \mathbf{U}_\ell = \begin{bmatrix} \mathbf{U}_{\ell-1} & u_\ell \end{bmatrix}.$$

We repeat this procedure until we have selected m indices.

Algorithm 1.5.1 (DEIM Algorithm).**Input:** $\{u_\ell\}_{\ell=1}^m \subset \mathbb{R}^n$ linearly independent**Output:** $\vec{\phi} = [\phi_1, \dots, \phi_m]^T \in \{1, \dots, n\}^m$ $\phi_1 \leftarrow \operatorname{argmax}_{1 \leq i \leq n} |u_1(i)|$ $\mathbf{U} \leftarrow [u_1], \quad \mathbf{P} \leftarrow [e_{\phi_1}], \quad \vec{\phi} \leftarrow [\phi_1]$ **for** $\ell = 2, \dots, m$ **do** Solve $(\mathbf{P}^T \mathbf{U})c = \mathbf{P}^T u_\ell$ for c $\mathbf{r} \leftarrow u_\ell - \mathbf{U}c$ $\phi_\ell \leftarrow \operatorname{argmax}_{1 \leq i \leq n} |\mathbf{r}(i)|$ $\mathbf{U} \leftarrow [\mathbf{U} \ u_\ell]$ $\mathbf{P} \leftarrow [\mathbf{P} \ e_{\phi_\ell}]$ $\vec{\phi} \leftarrow [\vec{\phi}^T \ \phi_\ell]^T$ **end for**

The pseudocode in Algorithm 1.5.1 summarizes the DEIM procedure. Notice that we have not specified how to obtain the basis vectors $\{u_\ell\}_{\ell=1}^m$. In our case, we use the POD-DEIM method, and later error estimates will justify this choice. As expected from their role in the SVD, the order of the basis vectors matters. A careful reader may notice that we have not explained why $\mathbf{P}^T \mathbf{U}_m$ is nonsingular at each iteration, but we will see later that this is indeed the case. Another point worth noting now is why the residual $r_{\ell-1}$ always has a nonzero entry: since the new vector u_ℓ is linearly independent of the previous ones, it cannot be represented perfectly by them, so the residual cannot be zero. Before going into error estimates, we present a numerical example to illustrate the DEIM method. The setup is summarized in Table 1.1.

Quantity	Value
Domain	$[0, 1]$
Function	$f(x; p) = (x - 0.5)^2 \sin(e^{11.2 p-0.5 } x(1-x))$
Training parameter interval	$[0.05, 1.0]$
Number of training parameters	40
Number of grid points	$n = 400$

Table 1.1: Setup for the one-dimensional DEIM experiment.

To visualize the function used in this example, we plot it for several parameter values.

Before applying the DEIM method, we first compute the POD basis, which requires choosing the number of basis vectors m . We experiment with different values of m to see how this choice affects the approximation, and then plot the function for previously unseen parameter values.

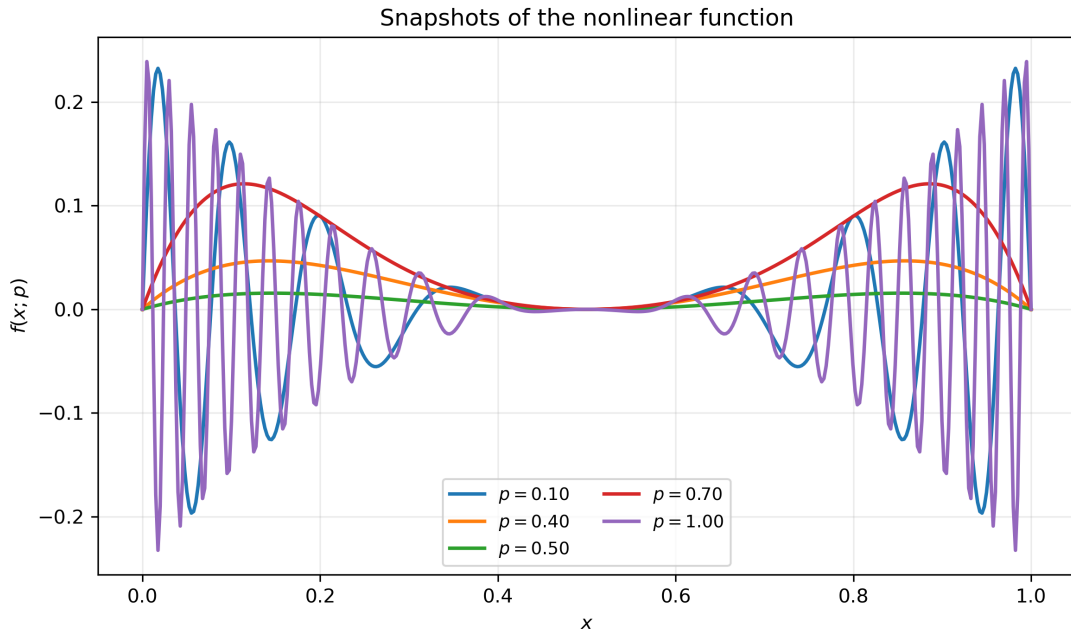


Figure 1.5: Snapshots of the one-dimensional nonlinear function for several parameter values.

The result can be seen in Figure 1.6. Since the approximation is lacking in some regions of the domain, we expect it to improve as the number of basis vectors m increases, as shown in Figure 1.7.

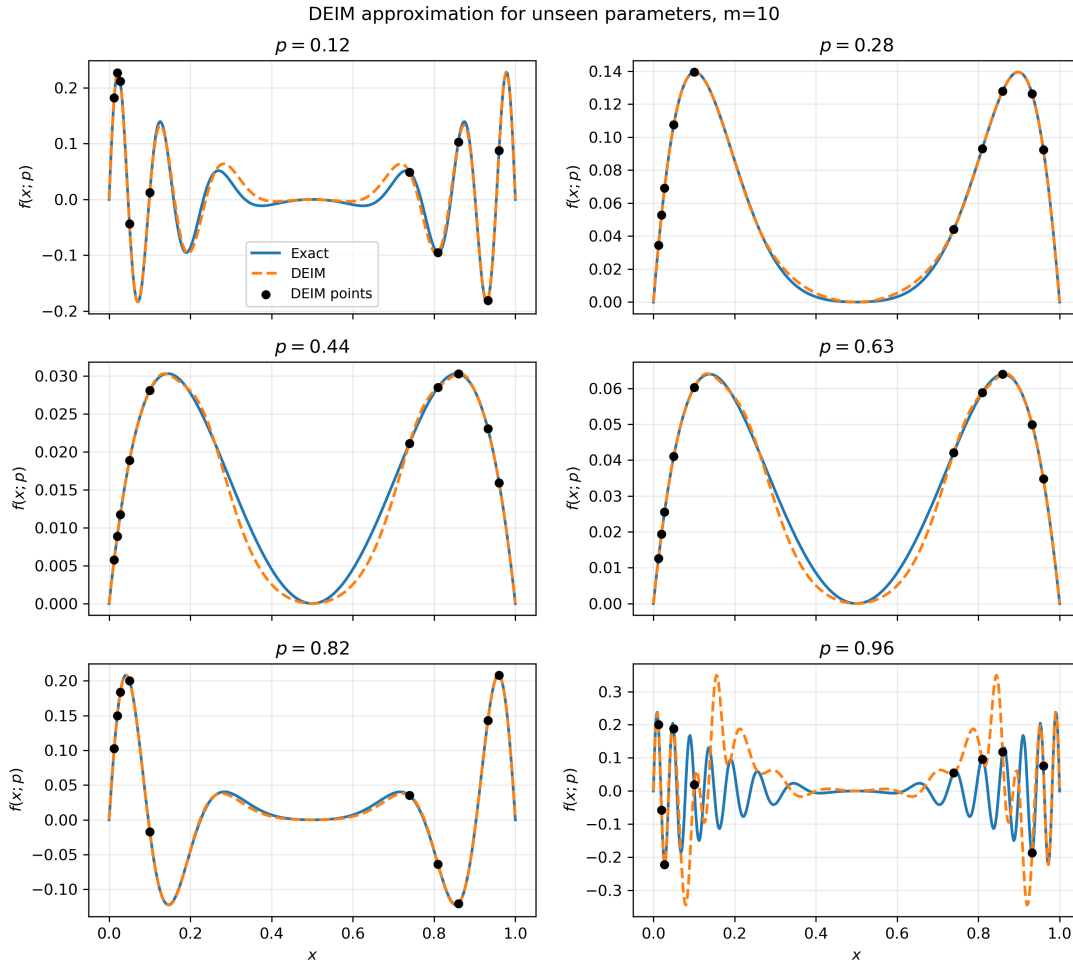


Figure 1.6: DEIM approximations of the one-dimensional nonlinear function at several unseen parameter values. The DEIM operator was constructed using a 10-dimensional POD basis.

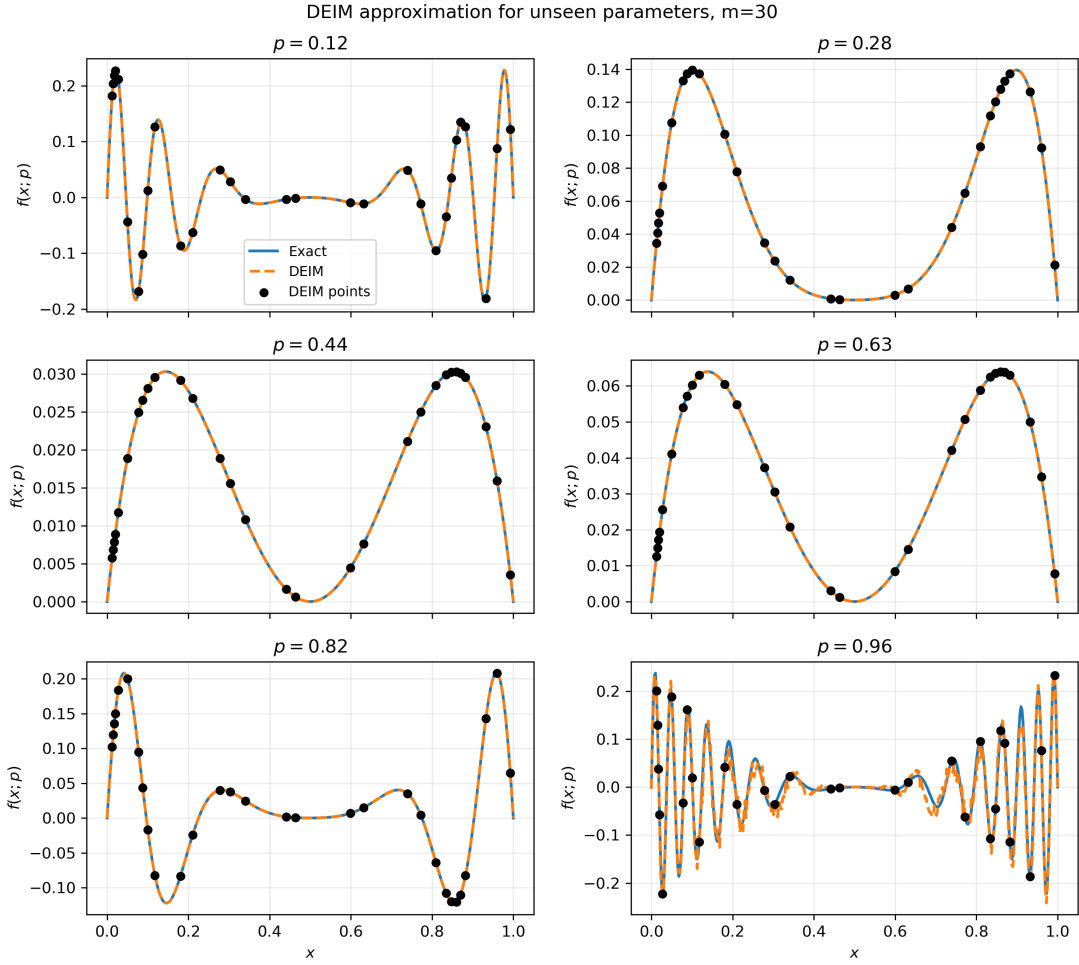


Figure 1.7: DEIM approximations of the one-dimensional nonlinear function at several unseen parameter values. The DEIM operator was constructed using a 30-dimensional POD basis.

1.6 Error Analysis for DEIM

We first record a norm identity for nontrivial projectors that will be used in the DEIM error estimate.

Lemma 1.6.1. *Let $P \in \mathbb{C}^{n \times n}$ be a projector such that $P \neq 0$ and $P \neq \mathbf{I}$. Then*

$$\|P\|_2 = \|\mathbf{I} - P\|_2.$$

Proof. Let $\mathcal{R} = \text{Ran}(P)$. Since P is non-trivial, both $\mathcal{R}$ and $\mathcal{R}^\perp$ are non-zero. Every vector in $\mathbb{C}^n$ can be written uniquely as (u, v) with $u \in \mathcal{R}$ and $v \in \mathcal{R}^\perp$. The projector acts as the identity on $\mathcal{R}$, while the image of every vector in $\mathcal{R}^\perp$ belongs to $\mathcal{R}$. Hence, with respect to the orthogonal decomposition $\mathbb{C}^n = \mathcal{R} \oplus \mathcal{R}^\perp$,

$$P = \begin{bmatrix} \mathbf{I} & A \\ 0 & 0 \end{bmatrix}, \quad \mathbf{I} - P = \begin{bmatrix} 0 & -A \\ 0 & \mathbf{I} \end{bmatrix},$$

where $A = P|_{\mathcal{R}^\perp} : \mathcal{R}^\perp \rightarrow \mathcal{R}$. Consequently,

$$P(u, v) = (u + Av, 0), \quad (\mathbf{I} - P)(u, v) = (-Av, v).$$

Set $a = \|A\|_2$. The triangle inequality followed by the Cauchy–Schwarz inequality in $\mathbb{R}^2$ gives

$$\begin{aligned} \|P(u, v)\|_2 &= \|u + Av\|_2 \\ &\leq \|u\|_2 + a\|v\|_2 \\ &\leq \sqrt{1 + a^2} \sqrt{\|u\|_2^2 + \|v\|_2^2}. \end{aligned}$$

Therefore,

$$\|P\|_2 \leq \sqrt{1 + a^2}.$$

Similarly,

$$\begin{aligned} \|(\mathbf{I} - P)(u, v)\|_2^2 &= \|Av\|_2^2 + \|v\|_2^2 \\ &\leq (1 + a^2)\|v\|_2^2 \\ &\leq (1 + a^2)(\|u\|_2^2 + \|v\|_2^2), \end{aligned}$$

and hence

$$\|\mathbf{I} - P\|_2 \leq \sqrt{1 + a^2}.$$

It remains to show that both upper bounds are attained. Since $\mathcal{R}^\perp$ is finite-dimensional, its unit sphere is compact. Thus, there exists a unit vector $v_0 \in \mathcal{R}^\perp$ such that $\|Av_0\|_2 = a$. If $a > 0$, set $w = Av_0/a$. Then $w \in \mathcal{R}$ and $\|w\|_2 = 1$. Taking $(u, v) = (w, av_0)$ yields

$$\frac{\|P(u, v)\|_2}{\|(u, v)\|_2} = \frac{1 + a^2}{\sqrt{1 + a^2}} = \sqrt{1 + a^2}.$$

On the other hand, taking $(u, v) = (0, v_0)$ gives

$$\frac{\|(\mathbf{I} - P)(u, v)\|_2}{\|(u, v)\|_2} = \sqrt{\|Av_0\|_2^2 + \|v_0\|_2^2} = \sqrt{1 + a^2}.$$

Therefore, $\|P\|_2 = \|\mathbf{I} - P\|_2 = \sqrt{1 + a^2}$ when $a > 0$. If $a = 0$, then P and $\mathbf{I} - P$ are non-zero orthogonal projectors, so both norms are equal to 1. This completes the proof. $\square$

Theorem 1.6.2. *Let $\mathbf{v} \in \mathbb{R}^n$ be an arbitrary vector and let $\{\mathbf{u}_1, \dots, \mathbf{u}_m\} \subset \mathbb{R}^n$ be an orthonormal set. Define*

$$\mathbf{U} = [\mathbf{u}_1 \quad \dots \quad \mathbf{u}_m] \in \mathbb{R}^{n \times m}.$$

Let

$$\mathbf{P} = [\mathbf{e}_{\varphi_1} \quad \dots \quad \mathbf{e}_{\varphi_m}] \in \mathbb{R}^{n \times m}$$

be the selection matrix generated by Algorithm 1.5.1 from the basis $\{\mathbf{u}_i\}_{i=1}^m$ and let $\widehat{\mathbf{v}}$ be the corresponding DEIM approximation

$$\widehat{\mathbf{v}} = \mathbf{U}(\mathbf{P}^T \mathbf{U})^{-1} \mathbf{P}^T \mathbf{v}.$$

Then, the approximation error satisfies

$$\|\mathbf{v} - \widehat{\mathbf{v}}\|_2 \leq \|(\mathbf{P}^T \mathbf{U})^{-1}\|_2 \|(\mathbf{I} - \mathbf{U} \mathbf{U}^T) \mathbf{v}\|_2. \quad (1.5)$$

Furthermore, the interpolation constant is bounded by

$$\|(\mathbf{P}^T \mathbf{U})^{-1}\|_2 \leq \frac{(1 + \sqrt{2n})^{m-1}}{|\mathbf{e}_{\varphi_1}^T \mathbf{u}_1|} = \frac{(1 + \sqrt{2n})^{m-1}}{\|\mathbf{u}_1\|_\infty}. \quad (1.6)$$

Proof. First we focus on the inequality (1.5). Define the DEIM projector

$$\mathbb{P} = \mathbf{U}(\mathbf{P}^T \mathbf{U})^{-1} \mathbf{P}^T.$$

Since we are approximating from $\text{Range}(\mathbf{U})$, the best approximation to $\mathbf{v}$ from that subspace is the orthogonal projection

$$\mathbf{v}_* = \mathbf{U} \mathbf{U}^T \mathbf{v},$$

with corresponding residual

$$\mathbf{r} = \mathbf{v} - \mathbf{v}_* = (\mathbf{I} - \mathbf{U}\mathbf{U}^T)\mathbf{v}.$$

Notice that

$$\mathbb{P}\mathbf{v}_* = (\mathbf{U}(\mathbf{P}^T\mathbf{U})^{-1}\mathbf{P}^T)\mathbf{U}\mathbf{U}^T\mathbf{v} = \mathbf{U}\mathbf{U}^T\mathbf{v} = \mathbf{v}_*.$$

This implies

$$\widehat{\mathbf{v}} = \mathbb{P}\mathbf{v} = \mathbb{P}\mathbf{v}_* + \mathbb{P}\mathbf{r} = \mathbf{v}_* + \mathbb{P}\mathbf{r}.$$

Therefore,

$$\|\mathbf{v} - \widehat{\mathbf{v}}\|_2 = \|(\mathbf{r} + \mathbf{v}_*) - (\mathbf{v}_* + \mathbb{P}\mathbf{r})\|_2 = \|(\mathbf{I} - \mathbb{P})\mathbf{r}\|_2 \leq \|\mathbf{I} - \mathbb{P}\|_2 \|\mathbf{r}\|_2.$$

To conclude (1.5), it remains to bound $\|\mathbf{I} - \mathbb{P}\|_2$. First note that $\mathbb{P}$ is a projector, since

$$\mathbb{P}^2 = \mathbf{U}(\mathbf{P}^T\mathbf{U})^{-1}\mathbf{P}^T\mathbf{U}(\mathbf{P}^T\mathbf{U})^{-1}\mathbf{P}^T = \mathbf{U}(\mathbf{P}^T\mathbf{U})^{-1}\mathbf{P}^T = \mathbb{P}.$$

If $\mathbb{P} = \mathbf{I}$, then $\|\mathbf{I} - \mathbb{P}\|_2 = 0$ and the desired bound is immediate. Otherwise, Lemma 1.6.1 gives

$$\|\mathbf{I} - \mathbb{P}\|_2 = \|\mathbb{P}\|_2.$$

Using the definition of $\mathbb{P}$ and submultiplicativity of the spectral norm, we obtain

$$\|\mathbf{I} - \mathbb{P}\|_2 = \|\mathbb{P}\|_2 = \|\mathbf{U}(\mathbf{P}^T\mathbf{U})^{-1}\mathbf{P}^T\|_2 \leq \|\mathbf{U}\|_2 \|(\mathbf{P}^T\mathbf{U})^{-1}\|_2 \|\mathbf{P}^T\|_2.$$

Since both $\mathbf{U}$ and $\mathbf{P}$ have orthonormal columns, we have $\|\mathbf{U}\|_2 = \|\mathbf{P}^T\|_2 = 1$. Hence

$$\|\mathbf{I} - \mathbb{P}\|_2 \leq \|(\mathbf{P}^T\mathbf{U})^{-1}\|_2.$$

Together with $\mathbf{r} = (\mathbf{I} - \mathbf{U}\mathbf{U}^T)\mathbf{v}$, this proves the inequality (1.5).

We now prove the bound (1.6). For $\ell \in \{1, \dots, m\}$, write

$$\begin{aligned} \overline{\mathbf{U}} &= [\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_{\ell-1}] \in \mathbb{R}^{n \times (\ell-1)}, & \overline{\mathbf{P}} &= [\mathbf{e}_{\varphi_1} \quad \cdots \quad \mathbf{e}_{\varphi_{\ell-1}}] \in \mathbb{R}^{n \times (\ell-1)}, \\ \mathbf{u} &= \mathbf{u}_\ell \in \mathbb{R}^n, & \mathbf{p} &= \mathbf{e}_{\varphi_\ell} \in \mathbb{R}^n, \\ \mathbf{U} &= [\overline{\mathbf{U}} \quad \mathbf{u}] \in \mathbb{R}^{n \times \ell}, & \mathbf{P} &= [\overline{\mathbf{P}} \quad \mathbf{p}] \in \mathbb{R}^{n \times \ell}. \end{aligned}$$

Define

$$\mathbf{M} = \mathbf{P}^T\mathbf{U}.$$

The matrices $\overline{\mathbf{U}}$ and $\overline{\mathbf{P}}$ are exactly the basis and selection matrices from the previous step of Algorithm 1.5.1, so we estimate $\|\mathbf{M}^{-1}\|_2$ by induction on ℓ . For $\ell = 1$, Algorithm 1.5.1 chooses φ_1 so that $|\mathbf{e}_{\varphi_1}^T \mathbf{u}_1| = \|\mathbf{u}_1\|_\infty$. Hence

$$\mathbf{P} = \mathbf{e}_{\varphi_1}, \quad \mathbf{U} = \mathbf{u}_1, \quad \mathbf{M}_1 = \mathbf{P}^T\mathbf{U} = \mathbf{e}_{\varphi_1}^T \mathbf{u}_1,$$

and therefore

$$\|\mathbf{M}_1^{-1}\|_2 = \frac{1}{|\mathbf{e}_{\varphi_1}^T \mathbf{u}_1|} = \frac{1}{\|\mathbf{u}_1\|_\infty}.$$

This is precisely (1.6) for $\ell = 1$. For the inductive step, let $\ell \geq 2$ and assume that $\overline{\mathbf{M}}$ is nonsingular. We have

$$\mathbf{M} = \begin{bmatrix} \overline{\mathbf{M}} & \overline{\mathbf{P}}^T \mathbf{u} \\ \mathbf{p}^T \overline{\mathbf{U}} & \mathbf{p}^T \mathbf{u} \end{bmatrix} = \begin{bmatrix} \overline{\mathbf{M}} & \mathbf{0} \\ \mathbf{a}^T & \rho \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{c} \\ \mathbf{0} & 1 \end{bmatrix}$$

where $\mathbf{a}^T = \mathbf{p}^T \overline{\mathbf{U}}$, $\rho = \mathbf{p}^T \mathbf{u} - \mathbf{a}^T \mathbf{c}$ and $\mathbf{c} = \overline{\mathbf{M}}^{-1} \overline{\mathbf{P}}^T \mathbf{u}$. Notice that $\rho = \mathbf{p}^T (\mathbf{u} - \overline{\mathbf{U}} \overline{\mathbf{M}}^{-1} \overline{\mathbf{P}}^T \mathbf{u})$ so $|\rho| = \|\mathbf{r}\|_\infty$ where $\mathbf{r}$ is the residual at iteration ℓ of Algorithm 1.5.1. Now, nonsingularity of $\overline{\mathbf{M}}$ implies

$$\begin{aligned} \mathbf{M}^{-1} &= \begin{bmatrix} \mathbf{I} & -\mathbf{c} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \overline{\mathbf{M}}^{-1} & \mathbf{0} \\ -\rho^{-1} \mathbf{a}^T \overline{\mathbf{M}}^{-1} & \rho^{-1} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{I} & -\mathbf{c} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ -\rho^{-1} \mathbf{a}^T & \rho^{-1} \end{bmatrix} \begin{bmatrix} \overline{\mathbf{M}}^{-1} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \\ &= \left\{ \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix} + \rho^{-1} \begin{bmatrix} \mathbf{c} \\ -1 \end{bmatrix} \begin{bmatrix} \mathbf{a}^T & -1 \end{bmatrix} \right\} \begin{bmatrix} \overline{\mathbf{M}}^{-1} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix}. \end{aligned}$$

Therefore, using submultiplicativity of the spectral norm,

$$\|\mathbf{M}^{-1}\|_2 \leq \left(\left\| \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix} \right\|_2 + |\rho|^{-1} \left\| \begin{bmatrix} \mathbf{c} \\ -1 \end{bmatrix} \begin{bmatrix} \mathbf{a}^T & -1 \end{bmatrix} \right\|_2 \right) \left\| \begin{bmatrix} \overline{\mathbf{M}}^{-1} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \right\|_2.$$

Now observe that

$$\begin{aligned} \left\| \begin{bmatrix} \mathbf{c} \\ -1 \end{bmatrix} \begin{bmatrix} \mathbf{a}^T & -1 \end{bmatrix} \right\|_2 &= \left\| \begin{bmatrix} \overline{\mathbf{U}} & \mathbf{u} \end{bmatrix} \begin{bmatrix} \mathbf{c} \\ -1 \end{bmatrix} \begin{bmatrix} \mathbf{a}^T & -1 \end{bmatrix} \right\|_2 \\ &\leq \|\overline{\mathbf{U}} \mathbf{c} - \mathbf{u}\|_2 \left\| \begin{bmatrix} \mathbf{a}^T & -1 \end{bmatrix} \right\|_2 \\ &\leq \sqrt{1 + \|\mathbf{a}\|_2^2} \sqrt{n} \|\overline{\mathbf{U}} \mathbf{c} - \mathbf{u}\|_\infty \\ &\leq \sqrt{2n} |\rho|. \end{aligned}$$

Here we used that $\begin{bmatrix} \overline{\mathbf{U}} & \mathbf{u} \end{bmatrix}$ has orthonormal columns, $\|\mathbf{a}\|_2 = \|\overline{\mathbf{U}}^T \mathbf{p}\|_2 \leq 1$, and $|\rho| = \|\mathbf{r}\|_\infty$

with $\mathbf{r} = \mathbf{u} - \overline{\mathbf{U}} \mathbf{c}$. Since $\left\| \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix} \right\|_2 = 1$ and $\left\| \begin{bmatrix} \overline{\mathbf{M}}^{-1} & \mathbf{0} \\ \mathbf{0} & 1 \end{bmatrix} \right\|_2 = \|\overline{\mathbf{M}}^{-1}\|_2$, we obtain

$$\|\mathbf{M}^{-1}\|_2 \leq (1 + \sqrt{2n}) \|\overline{\mathbf{M}}^{-1}\|_2.$$

Applying this estimate recursively from $\ell = m$ down to $\ell = 1$ gives

$$\|(\mathbf{P}^T \mathbf{U})^{-1}\|_2 \leq (1 + \sqrt{2n})^{m-1} \|\mathbf{M}_1^{-1}\|_2 = \frac{(1 + \sqrt{2n})^{m-1}}{|\mathbf{e}_{\varphi_1}^T \mathbf{u}_1|} = \frac{(1 + \sqrt{2n})^{m-1}}{\|\mathbf{u}_1\|_\infty},$$

which proves (1.6). □

1.7 QDEIM

The error estimate (1.5) shows that, once the approximation space $\text{Range}(\mathbf{U}_m)$ has been fixed, the quality of the DEIM approximation is influenced by the interpolation constant

$$c(\mathbf{P}, \mathbf{U}_m) := \|(\mathbf{P}^T \mathbf{U}_m)^{-1}\|_2.$$

The POD construction controls how well $\text{Range}(\mathbf{U}_m)$ represents the available snapshots, whereas $\mathbf{P}$ controls how reliably a vector in this space can be recovered from its selected entries. The remaining problem is therefore to select m rows of $\mathbf{U}_m$ that form a nonsingular and preferably well-conditioned submatrix.

The original DEIM algorithm approaches this problem sequentially. It processes the ordered basis vectors $u_1, \dots, u_m$ one at a time and selects a new index from the residual of the current basis vector. QDEIM, introduced by Drmač and Gugercin [9], replaces this sequential selection by a column-pivoted QR factorization. In this way all basis vectors participate in the selection simultaneously, and the resulting indices depend on the approximation subspace rather than on a particular orthonormal basis used to represent it.

1.7.1 From row selection to column pivoting

Let

$$\mathbf{U}_m \in \mathbb{R}^{n \times m}, \quad \mathbf{U}_m^T \mathbf{U}_m = \mathbf{I}_m, \quad m < n.$$

Selecting rows of $\mathbf{U}_m$ is equivalent to selecting columns of the wide matrix $\mathbf{U}_m^T \in \mathbb{R}^{m \times n}$. This observation allows us to use the column-pivoted QR factorization introduced earlier. We compute

$$\mathbf{U}_m^T \Pi = Q \begin{bmatrix} R_{11} & R_{12} \end{bmatrix}, \quad (1.7)$$

where $Q \in \mathbb{R}^{m \times m}$ is orthogonal, $\Pi \in \mathbb{R}^{n \times n}$ is a permutation matrix, and $R_{11} \in \mathbb{R}^{m \times m}$ is upper triangular. Since $\mathbf{U}_m^T$ has full row rank, the first m pivots are nonzero and R_{11} is nonsingular.

Write the permutation matrix in terms of its columns as

$$\Pi = \begin{bmatrix} e_{p_1} & \cdots & e_{p_m} & e_{p_{m+1}} & \cdots & e_{p_n} \end{bmatrix}.$$

QDEIM retains the first m pivot indices and defines the selection matrix

$$\mathbf{P} = \begin{bmatrix} e_{p_1} & \cdots & e_{p_m} \end{bmatrix} = \Pi(:, 1 : m).$$

Taking the first m columns in (1.7) gives

$$\mathbf{U}_m^T \mathbf{P} = QR_{11},$$

and taking the transpose yields

$$\mathbf{P}^T \mathbf{U}_m = R_{11}^T Q^T.$$

Because Q preserves the norm, it follows that

$$\|(\mathbf{P}^T \mathbf{U}_m)^{-1}\|_2 = \|R_{11}^{-1}\|_2. \quad (1.8)$$

Thus QDEIM seeks a leading triangular block R_{11} whose inverse has moderate norm.

The pivoting rule also has a useful geometric interpretation. Each column of $\mathbf{U}_m^T$ is a row of $\mathbf{U}_m$. At the first step, column-pivoted QR selects a column of largest Euclidean norm, so the first QDEIM index corresponds to a row of $\mathbf{U}_m$ with largest row norm. After $\ell - 1$ steps, if $Q_{\ell-1}$ contains the orthonormal directions already constructed, the next pivot is chosen according to

$$p_\ell \in \underset{j \notin \{p_1, \dots, p_{\ell-1}\}}{\operatorname{argmax}} \left\| (\mathbf{I} - Q_{\ell-1} Q_{\ell-1}^T) \mathbf{U}_m^T e_j \right\|_2.$$

In other words, QDEIM chooses the remaining row containing the largest component that is not already represented by the previously selected rows. The role of pivoting is therefore not merely to find independent rows, but to greedily avoid choices that add very little new information.

1.7.2 The QDEIM approximation

The complete index-selection procedure is summarized below.

Algorithm 1.7.1 (QDEIM index selection).

Input: $\mathbf{U}_m \in \mathbb{R}^{n \times m}$ with $\mathbf{U}_m^T \mathbf{U}_m = \mathbf{I}_m$

Output: interpolation indices $p_1, \dots, p_m$ and selection matrix $\mathbf{P}$

Compute $\mathbf{U}_m^T \Pi = Q \begin{bmatrix} R_{11} & R_{12} \end{bmatrix}$ by column-pivoted QR

Write $\Pi = \begin{bmatrix} e_{p_1} & \cdots & e_{p_n} \end{bmatrix}$

$\mathbf{P} \leftarrow \begin{bmatrix} e_{p_1} & \cdots & e_{p_m} \end{bmatrix}$

Once $\mathbf{P}$ has been selected, QDEIM uses exactly the same interpolation formula as DEIM:

$$\widehat{f}(p) = \mathbf{U}_m(\mathbf{P}^T \mathbf{U}_m)^{-1} \mathbf{P}^T f(p). \quad (1.9)$$

Thus, QDEIM is not a different approximation ansatz; it is a different rule for constructing the selection matrix. In particular, (1.9) interpolates $f(p)$ at the selected entries and is exact whenever $f(p) \in \text{Range}(\mathbf{U}_m)$.

The matrix

$$\mathbf{D}_Q = \mathbf{U}_m(\mathbf{P}^T \mathbf{U}_m)^{-1}$$

can be precomputed, so for each new parameter value only the selected entries $\mathbf{P}^T f(p)$ are required. As for standard DEIM, this reduces the online cost only if those entries can be evaluated without assembling the full vector $f(p)$.

1.7.3 Basis invariance and the QDEIM bound

An important distinction between DEIM and QDEIM concerns a change of basis. Suppose that the same approximation space is represented by

$$\widetilde{\mathbf{U}}_m = \mathbf{U}_m \Omega, \quad \Omega^T \Omega = \mathbf{I}_m.$$

Then

$$\widetilde{\mathbf{U}}_m^T = \Omega^T \mathbf{U}_m^T.$$

Left multiplication by the orthogonal matrix Ω^T preserves the column norms and all residual column norms used by the Businger–Golub pivoting rule. Consequently, in exact arithmetic QDEIM produces the same pivot indices for $\mathbf{U}_m$ and $\widetilde{\mathbf{U}}_m$, provided that ties are resolved in the same way. Moreover,

$$\begin{aligned} \widetilde{\mathbf{U}}_m(\mathbf{P}^T \widetilde{\mathbf{U}}_m)^{-1} \mathbf{P}^T &= \mathbf{U}_m \Omega (\mathbf{P}^T \mathbf{U}_m \Omega)^{-1} \mathbf{P}^T \\ &= \mathbf{U}_m (\mathbf{P}^T \mathbf{U}_m)^{-1} \mathbf{P}^T. \end{aligned}$$

Hence, the selected indices and the interpolation operator are determined by $\text{Range}(\mathbf{U}_m)$, not by its particular orthonormal basis. The original DEIM selection does not generally have this invariance because it processes the basis vectors in a prescribed order.

The QDEIM analysis gives the a priori estimate

$$\|(\mathbf{P}^T \mathbf{U}_m)^{-1}\|_2 \leq \sqrt{n-m+1} \frac{\sqrt{4^m + 6m - 1}}{3}. \quad (1.10)$$

Combining (1.10) with the general DEIM error estimate (1.5) yields

$$\|f(p) - \widehat{f}(p)\|_2 \leq \sqrt{n-m+1} \frac{\sqrt{4^m + 6m - 1}}{3} \|(\mathbf{I} - \mathbf{U}_m \mathbf{U}_m^T) f(p)\|_2.$$

Although pessimistic, this estimate depends on the ambient dimension n much more mildly than the original DEIM bound (1.6). Column-pivoted QR costs $\mathcal{O}(nm^2)$, comparable to the original DEIM selection, and the following chapter compares the two rules numerically.

1.8 Restricted interpolation indices

The DEIM and QDEIM selection procedures considered so far allow every coordinate to be selected. In applications this freedom may be unrealistic. Some spatial locations may be inaccessible, measurements may be unavailable or unreliable in part of the domain, or particular entries of a nonlinear vector may be too expensive to evaluate. The restricted and randomized QDEIM framework already notes that rows associated with corrupted or missing spatial information may need to be avoided [9]. Related QR-based sensor-selection methods also incorporate inaccessible locations and location-dependent costs [8].

There are two interpretations of the selected entries which should be distinguished. In hyper-reduction, $\mathbf{P}^T f$ contains selected entries of a nonlinear vector which are evaluated during the online stage. In state reconstruction, the same algebra describes physical measurements or sensors. The following construction applies to both situations.

1.8.1 The admissible index set

Let

$$\mathbf{U}_m \in \mathbb{R}^{n \times m}, \quad \mathbf{U}_m^T \mathbf{U}_m = \mathbf{I}_m,$$

be the approximation basis, and suppose that only the indices in

$$\mathcal{A} = \{a_1, \dots, a_q\} \subseteq \{1, \dots, n\}$$

are admissible. Introduce the matrix

$$\mathbf{E}_{\mathcal{A}} = \begin{bmatrix} e_{a_1} & \cdots & e_{a_q} \end{bmatrix} \in \mathbb{R}^{n \times q}.$$

The matrix

$$\mathbf{U}_{\mathcal{A}} = \mathbf{E}_{\mathcal{A}}^T \mathbf{U}_m \in \mathbb{R}^{q \times m} \tag{1.11}$$

is obtained by retaining only the admissible rows of $\mathbf{U}_m$.

Every selection matrix whose indices belong to $\mathcal{A}$ has the form

$$\mathbf{P} = \mathbf{E}_{\mathcal{A}} \widehat{\mathbf{P}}, \tag{1.12}$$

where $\widehat{\mathbf{P}} \in \mathbb{R}^{q \times m}$ selects m coordinates inside the admissible set. Consequently,

$$\mathbf{P}^T \mathbf{U}_m = \widehat{\mathbf{P}}^T \mathbf{U}_{\mathcal{A}}. \tag{1.13}$$

Thus the restricted selection problem is equivalent to finding a nonsingular $m \times m$ row submatrix of $\mathbf{U}_{\mathcal{A}}$.

Theorem 1.8.1. *There exists a selection matrix $\mathbf{P}$ containing only indices from $\mathcal{A}$ for which $\mathbf{P}^T \mathbf{U}_m$ is nonsingular if and only if*

$$\text{rank}(\mathbf{E}_{\mathcal{A}}^T \mathbf{U}_m) = m. \quad (1.14)$$

Proof. Suppose first that an admissible $\mathbf{P}$ exists. By (1.13), $\mathbf{P}^T \mathbf{U}_m$ is a row submatrix of $\mathbf{U}_{\mathcal{A}}$. Since this submatrix has rank m , $\mathbf{U}_{\mathcal{A}} = \mathbf{E}_{\mathcal{A}}^T \mathbf{U}_m$ also has rank m .

Conversely, if $\mathbf{U}_{\mathcal{A}}$ has rank m , then it contains m linearly independent rows. Selecting these rows defines $\widehat{\mathbf{P}}$ and hence an admissible $\mathbf{P}$ for which $\mathbf{P}^T \mathbf{U}_m$ is nonsingular. $\square$

Condition (1.14) requires $q \geq m$. More importantly, it says that every direction in $\text{Range}(\mathbf{U}_m)$ must be visible from the admissible coordinates. If the condition fails, there is a nonzero vector $c \in \mathbb{R}^m$ such that

$$\mathbf{E}_{\mathcal{A}}^T \mathbf{U}_m c = 0.$$

The nonzero basis combination $\mathbf{U}_m c$ is then zero at every admissible coordinate and cannot be distinguished from the zero vector using the available information.

1.8.2 Restricted QDEIM

Selecting rows of $\mathbf{U}_{\mathcal{A}}$ is equivalent to selecting columns of

$$\mathbf{U}_{\mathcal{A}}^T = \mathbf{U}_m^T \mathbf{E}_{\mathcal{A}} \in \mathbb{R}^{m \times q}.$$

Assuming (1.14), compute the column-pivoted QR factorization

$$\mathbf{U}_m^T \mathbf{E}_{\mathcal{A}} \Pi_{\mathcal{A}} = Q \begin{bmatrix} R_{11} & R_{12} \end{bmatrix}, \quad (1.15)$$

where $R_{11} \in \mathbb{R}^{m \times m}$ is nonsingular. The first m pivots select admissible coordinates:

$$\widehat{\mathbf{P}} = \Pi_{\mathcal{A}}(:, 1 : m), \quad \mathbf{P} = \mathbf{E}_{\mathcal{A}} \widehat{\mathbf{P}}.$$

Indeed,

$$\mathbf{U}_m^T \mathbf{P} = Q R_{11},$$

and therefore

$$\mathbf{P}^T \mathbf{U}_m = R_{11}^T Q^T, \quad \|(\mathbf{P}^T \mathbf{U}_m)^{-1}\|_2 = \|R_{11}^{-1}\|_2. \quad (1.16)$$

Algorithm 1.8.2 (Restricted QDEIM index selection).**Input:** $\mathbf{U}_m \in \mathbb{R}^{n \times m}$ and $\mathcal{A} = \{a_1, \dots, a_q\}$ **Output:** admissible indices $p_1, \dots, p_m$ and $\mathbf{P}$ Form $\mathbf{E}_{\mathcal{A}} = \begin{bmatrix} e_{a_1} & \cdots & e_{a_q} \end{bmatrix}$ Verify that $\text{rank}(\mathbf{E}_{\mathcal{A}}^T \mathbf{U}_m) = m$ Compute $\mathbf{U}_m^T \mathbf{E}_{\mathcal{A}} \Pi_{\mathcal{A}} = \mathcal{Q} \begin{bmatrix} R_{11} & R_{12} \end{bmatrix}$ $\widehat{\mathbf{P}} \leftarrow \Pi_{\mathcal{A}}(:, 1 : m)$ $\mathbf{P} \leftarrow \mathbf{E}_{\mathcal{A}} \widehat{\mathbf{P}}$

Once $\mathbf{P}$ has been obtained, the approximation has the usual form

$$\widehat{f} = \mathbf{U}_m (\mathbf{P}^T \mathbf{U}_m)^{-1} \mathbf{P}^T f. \quad (1.17)$$

The restriction changes the set over which the indices are chosen, but it does not change the interpolation formula.

1.8.3 Conditioning and loss of information

The full-rank condition guarantees only that restricted interpolation is possible. Its stability is governed by

$$\|(\mathbf{P}^T \mathbf{U}_m)^{-1}\|_2 = \frac{1}{\sigma_{\min}(\mathbf{P}^T \mathbf{U}_m)}.$$

If the admissible region sees one basis direction only weakly, the smallest singular value may be close to zero even though the matrix has full rank. The general DEIM estimate remains

$$\|f - \widehat{f}\|_2 \leq \|(\mathbf{P}^T \mathbf{U}_m)^{-1}\|_2 \|(\mathbf{I} - \mathbf{U}_m \mathbf{U}_m^T) f\|_2. \quad (1.18)$$

The basis error on the right-hand side is unchanged. Restricting the indices can affect only the interpolation constant.

To make this precise, let $\mathcal{S}$ denote all selections of m indices and let $\mathcal{S}_{\mathcal{A}}$ denote those contained in $\mathcal{A}$. Since $\mathcal{S}_{\mathcal{A}} \subseteq \mathcal{S}$,

$$\min_{\mathbf{P} \in \mathcal{S}_{\mathcal{A}}} \|(\mathbf{P}^T \mathbf{U}_m)^{-1}\|_2 \geq \min_{\mathbf{P} \in \mathcal{S}} \|(\mathbf{P}^T \mathbf{U}_m)^{-1}\|_2.$$

Thus an admissibility restriction cannot improve the best achievable interpolation constant, although a particular restricted selection can still outperform a particular unrestricted greedy selection.

The a priori QDEIM bound (1.10) was derived using all rows of an orthonormal matrix. It should therefore not be quoted unchanged for (1.15), because $\mathbf{E}_{\mathcal{A}}^T \mathbf{U}_m$ generally no longer has orthonormal columns. In the restricted setting, the computed quantity $\|\mathbf{R}_{11}^{-1}\|_2$ and the singular values of the admissible basis provide direct diagnostics of stability.

1.8.4 Hard restrictions, costs, and oversampling

A hard restriction declares some locations completely unavailable. A softer model assigns a nonnegative cost η_j to every candidate location and balances the pivot residual against this cost. Cost-modified QR selection of this type is studied in [8]. Completely inaccessible locations correspond to infinite cost and are handled by removing them before the QR factorization, which gives exactly (1.15).

For physical measurements, selecting exactly m coordinates may be sensitive to noise. If $s > m$ admissible coordinates are selected with

$$\mathbf{S} \in \mathbb{R}^{n \times s}, \quad \text{rank}(\mathbf{S}^T \mathbf{U}_m) = m,$$

the coefficients can instead be determined by least squares:

$$c = (\mathbf{S}^T \mathbf{U}_m)^\dagger \mathbf{S}^T f, \quad \widehat{f} = \mathbf{U}_m (\mathbf{S}^T \mathbf{U}_m)^\dagger \mathbf{S}^T f.$$

This is the gappy POD form of the approximation. Oversampling and its stability in the presence of perturbed measurements are studied in [11]. The sensitivity to measurement noise is controlled by

$$\|(\mathbf{S}^T \mathbf{U}_m)^\dagger\|_2 = \frac{1}{\sigma_{\min}(\mathbf{S}^T \mathbf{U}_m)}.$$

Additional admissible samples can therefore compensate, at least partly, for the loss of favorable locations caused by the restriction.

1.9 Taking a Step Back: What Does the Reduced Model Solve?

This section is a brief digression from the main topic of this thesis. When working with ordinary or partial differential equations, it is important that the corresponding initial- or boundary-value problem be well posed, meaning that a solution exists, is unique, and depends continuously on the data. The methods developed in this thesis are designed to reduce the computational cost at different stages of the model reduction process, but it is important to note that we are almost always working with a reduced model, not the original full-order one. We illustrate this distinction with a simple example.

Suppose we have a parameterized ordinary differential equation (ODE) of the form

$$\mathbf{M}\dot{x}(t; \mu) = f(x(t; \mu), t; \mu), \quad x(0; \mu) = x_0(\mu), \quad x(t; \mu) \in \mathbb{R}^N. \quad (1.19)$$

This is the full-order initial-value problem. We will always implicitly assume that it is well posed on the time interval of interest for every parameter $\mu \in \mathcal{D}$.

We do not assume that the exact solution belongs to a low-dimensional space. Instead, we choose a reduced basis $\Phi \in \mathbb{R}^{N \times n_r}$, where $n_r \ll N$, and look for an approximation of the form

$$x_r(t; \mu) = x_{\text{ref}} + \Phi \hat{x}(t; \mu), \quad (1.20)$$

where $\hat{x}(t; \mu) \in \mathbb{R}^{n_r}$ contains the reduced coordinates. For simplicity, we take $x_{\text{ref}} = x_0(\mu)$ and $\hat{x}(0; \mu) = 0$, so the initial condition is satisfied exactly. At this point, $\hat{x}$ is simply an unknown differentiable function.

Substituting x_r into the full-order equation does not generally produce an equality. The difference between its two sides is the residual

$$r(t; \mu) = \mathbf{M}\Phi \dot{\hat{x}}(t; \mu) - f(x_{\text{ref}} + \Phi \hat{x}(t; \mu), t; \mu). \quad (1.21)$$

It measures how much the approximate trajectory fails to satisfy the full-order equation. Requiring $r = 0$ would give N equations for only n_r reduced coordinates, so this is generally not possible. The Galerkin method instead requires the residual to be orthogonal to the reduced space:

$$\Phi^T r(t; \mu) = 0. \quad (1.22)$$

Using (1.21), this gives the reduced system

$$\Phi^T \mathbf{M} \Phi \dot{\hat{x}}(t; \mu) = \Phi^T f(x_{\text{ref}} + \Phi \hat{x}(t; \mu), t; \mu), \quad \hat{x}(0; \mu) = 0. \quad (1.23)$$

If $\Phi^T \mathbf{M} \Phi$ is nonsingular, the system can be written as an ODE for $\hat{x}$. Under the usual regularity assumptions on f , this reduced initial-value problem also has a unique solution.

The solution of (1.23) is not, in general, the solution of the full-order problem (1.19). It defines an approximate trajectory through (1.20), and only the projected residual is zero. If the exact solution happens to belong to $x_{\text{ref}} + \text{Range}(\Phi)$ for all t , then its reduced coordinates satisfy (1.23). If the reduced problem has a unique solution, the Galerkin model reproduces the full-order solution in this special case. The existence of the full-order solution alone does not imply this.

DEIM introduces one more approximation. With the DEIM operator

$$\mathcal{D}_P = \mathbf{U}_m(\mathbf{P}^T \mathbf{U}_m)^{-1} \mathbf{P}^T, \quad (1.24)$$

we replace the nonlinear term by

$$f(x_r, t; \mu) \approx \mathcal{D}_P f(x_r, t; \mu).$$

The resulting hyper-reduced system is

$$\Phi^T \mathbf{M} \Phi \dot{\hat{x}}_D(t; \mu) = \Phi^T \mathcal{D}_P f(x_{\text{ref}} + \Phi \hat{x}_D(t; \mu), t; \mu), \quad \hat{x}_D(0; \mu) = 0. \quad (1.25)$$

We use a different symbol $\hat{x}_D$ because the Galerkin and DEIM systems generally have different solutions. Thus the full-order model, the Galerkin model, and the DEIM model are three different initial-value problems. Their solutions agree only in special cases; otherwise, their closeness must be studied using error estimates or numerical experiments.

In the usual POD–DEIM construction, Φ is obtained from solution snapshots, while $\mathbf{U}_m$ is obtained from nonlinear snapshots. The SNS method considered next changes how the second basis is constructed, but it does not change the distinction between the full-order and reduced problems.

1.10 Solution-Based Nonlinear Subspace Method - SNS

The construction of a POD–DEIM reduced model usually requires two separate collections of snapshots. Solution snapshots are compressed to obtain the state basis, while snapshots of the nonlinear term are compressed to obtain the DEIM basis which comes down to computing the SVD decomposition of a large matrix. For many full-order models, storing and compressing the second snapshot matrix can constitute a substantial part of the offline cost. The solution-based nonlinear subspace (SNS) method of Choi, Coombs, and Anderson [7] avoids this second construction by deriving a basis for the nonlinear term directly from the solution basis. Therefore, the procedure contains only one collection of snapshots and one SVD computation.

SNS and QDEIM pursue the same broader goal of making hyper-reduction more efficient, but they modify different parts of its offline workflow. QDEIM changes how interpolation indices are selected once a nonlinear basis is available. SNS instead changes how that nonlinear basis is constructed. In this section the usual DEIM selection procedure is retained; only its input basis is replaced by an SNS basis.

1.10.1 The nonlinear-snapshot subspace

Consider the parameter-dependent full-order model

$$\mathbf{M}\dot{x}(t; \mu) = f(x(t; \mu), t; \mu), \quad x(0; \mu) = x_0(\mu), \quad (1.26)$$

where $x(t; \mu) \in \mathbb{R}^n$ and $\mathbf{M} \in \mathbb{R}^{n \times n}$ is nonsingular. To see the relation between solution snapshots and nonlinear snapshots, first discretize (1.26) by the forward Euler method. With $t_k = t_{k-1} + \Delta t$, $x^k = x(t_k; \mu)$, and $f^k = f(x^k, t_k; \mu)$, the discrete equations are

$$\mathbf{M}(x^k - x^{k-1}) = \Delta t f^{k-1}.$$

Consequently,

$$f^{k-1} = \frac{1}{\Delta t} \mathbf{M}x^k - \frac{1}{\Delta t} \mathbf{M}x^{k-1}, \quad (1.27)$$

so every nonlinear snapshot is a linear combination of two $\mathbf{M}$ -scaled solution snapshots. Along the complete trajectory this gives the exact inclusion

$$\text{span}\{f^0, \dots, f^{N_t-1}\} \subseteq \text{span}\{\mathbf{M}x^0, \dots, \mathbf{M}x^{N_t}\}. \quad (1.28)$$

The same observation follows from backward Euler. In that case,

$$\mathbf{M}(x^k - x^{k-1}) = \Delta t f^k,$$

and hence

$$\text{span}\{f^1, \dots, f^{N_t}\} \subseteq \text{span}\{\mathbf{M}x^0, \dots, \mathbf{M}x^{N_t}\}. \quad (1.29)$$

The difference between (1.28) and (1.29) is only the time level at which the right-hand side is evaluated. Similar relations can be derived for other linear multistep and Runge–Kutta methods.

If

$$\mathbf{X} = \begin{bmatrix} x^0 & x^1 & \dots & x^{N_t} \end{bmatrix}$$

is the solution snapshot matrix and $\mathbf{F}$ contains the corresponding nonlinear snapshots, the preceding inclusions can be written compactly as

$$\text{Range}(\mathbf{F}) \subseteq \text{Range}(\mathbf{MX}). \quad (1.30)$$

For several training parameters, the same result holds for each trajectory and therefore also after their snapshot matrices have been concatenated.

Relation (1.30) is an algebraic statement about snapshots satisfying the chosen time-discrete equations. It does not claim that $f(x, t; \mu)$ belongs to this subspace for every possible state, time, and parameter. Moreover, after the solution snapshots have been compressed, the truncated state basis need not contain every direction required to represent the nonlinear snapshots. This distinction motivates the use of an extended solution basis.

1.10.2 The conforming-subspace condition

Let

$$x(t; \mu) \approx x_0(\mu) + \Phi \widehat{x}(t; \mu), \quad \Phi \in \mathbb{R}^{n \times n_s}, \quad \Phi^T \Phi = \mathbf{I}_{n_s},$$

be the reduced state approximation. Before projection, substitution in (1.26) gives

$$\mathbf{M} \Phi \widehat{x} = f(x_0 + \Phi \widehat{x}, t; \mu). \quad (1.31)$$

The left-hand side of (1.31) always belongs to $\text{Range}(\mathbf{M}\Phi)$. If the right-hand side is approximated by DEIM using a nonlinear basis Φ_f , its approximation belongs to $\text{Range}(\Phi_f)$. This leads to the conforming-subspace condition

$$\boxed{\text{Range}(\mathbf{M}\Phi) \subseteq \text{Range}(\Phi_f)}. \quad (1.32)$$

It ensures that all directions in which the reduced state can evolve are represented in the nonlinear approximation space. The condition is a compatibility requirement between the two spaces; it does not by itself make the unprojected, overdetermined equation exactly solvable for every nonlinear vector.

The smallest direct SNS choice is

$$\boxed{\Phi_f = \mathbf{M}\Phi}. \quad (1.33)$$

In this case the two ranges in (1.32) are equal, and the nonlinear basis has the same dimension as the state basis. This construction is inexpensive, but it may be restrictive: although (1.30) holds for the complete solution snapshot space, it does not imply

$$\text{Range}(\mathbf{F}) \subseteq \text{Range}(\mathbf{M}\Phi)$$

after POD truncation.

The nonlinear space can be enriched without increasing the number of reduced state variables. Let

$$\Phi_e = [\Phi \quad \Phi_E] \in \mathbb{R}^{n \times n_e}, \quad n_e > n_s,$$

contain additional left singular vectors of the same solution snapshot matrix, with $\Phi^T \Phi_E = 0$. SNS then defines

$$\boxed{\Phi_f = \mathbf{M}\Phi_e}. \quad (1.34)$$

Since

$$\text{Range}(\Phi) \subseteq \text{Range}(\Phi_e),$$

multiplication by the nonsingular matrix $\mathbf{M}$ gives

$$\text{Range}(\mathbf{M}\Phi) \subseteq \text{Range}(\mathbf{M}\Phi_e).$$

Thus (1.34) satisfies the conforming-subspace condition exactly. The state basis Φ still determines the reduced dimension n_s , while n_e controls the richness of the nonlinear approximation. No separate nonlinear snapshot matrix or nonlinear-snapshot SVD is required.

1.10.3 The DEIM–SNS reduced model

Let

$$\mathbf{P} = \begin{bmatrix} e_{p_1} & \cdots & e_{p_{n_e}} \end{bmatrix} \in \mathbb{R}^{n \times n_e}$$

be the selection matrix obtained by applying the DEIM index-selection procedure to the SNS nonlinear basis. Using $\Phi_f = \mathbf{M}\Phi_e$, the nonlinear term is approximated by

$$f \approx \mathbf{M}\Phi_e(\mathbf{P}^T\mathbf{M}\Phi_e)^{-1}\mathbf{P}^T f. \quad (1.35)$$

After inserting (1.35) into (1.31) and applying Galerkin projection, one obtains

$$\Phi^T\mathbf{M}\Phi\hat{\dot{x}} = \Phi^T\mathbf{M}\Phi_e(\mathbf{P}^T\mathbf{M}\Phi_e)^{-1}\mathbf{P}^T f(x_0 + \Phi\hat{x}, t; \mu).$$

Assuming that $\Phi^T\mathbf{M}\Phi$ is nonsingular and writing $\Phi_e = [\Phi \ \Phi_E]$, the reduced system can be expressed as

$$\begin{aligned} \hat{\dot{x}} = & \begin{bmatrix} \mathbf{I}_{n_s} & (\Phi^T\mathbf{M}\Phi)^{-1}\Phi^T\mathbf{M}\Phi_E \end{bmatrix} \\ & \cdot (\mathbf{P}^T\mathbf{M}\Phi_e)^{-1}\mathbf{P}^T f(x_0 + \Phi\hat{x}, t; \mu). \end{aligned} \quad (1.36)$$

For the basic choice $\Phi_e = \Phi$, this simplifies to

$$\hat{\dot{x}} = (\mathbf{P}^T\mathbf{M}\Phi)^{-1}\mathbf{P}^T f(x_0 + \Phi\hat{x}, t; \mu). \quad (1.37)$$

As in ordinary DEIM, an online reduction is obtained only when the selected entries $\mathbf{P}^T f$ can be evaluated without first assembling the full n -dimensional vector.

1.10.4 Orthonormalizing the SNS basis

Although Φ_e has orthonormal columns, $\mathbf{M}\Phi_e$ generally does not, since

$$(\mathbf{M}\Phi_e)^T(\mathbf{M}\Phi_e) = \Phi_e^T\mathbf{M}^T\mathbf{M}\Phi_e.$$

Because $\mathbf{M}$ is nonsingular, $\mathbf{M}\Phi_e$ has full column rank and admits an economic QR factorization

$$\mathbf{M}\Phi_e = \mathbf{Q}_f\mathbf{R}_f, \quad \mathbf{Q}_f^T\mathbf{Q}_f = \mathbf{I}_{n_e}. \quad (1.38)$$

The two matrices generate the same subspace:

$$\text{Range}(\mathbf{Q}_f) = \text{Range}(\mathbf{M}\Phi_e).$$

The SNS paper therefore recommends applying the sampling procedure to the orthonormal basis $\mathbf{Q}_f$. For a fixed selection matrix $\mathbf{P}$, this change of basis does not change the interpolation projector, because

$$\begin{aligned} \mathbf{M}\Phi_e(\mathbf{P}^T\mathbf{M}\Phi_e)^{-1}\mathbf{P}^T &= \mathbf{Q}_f\mathbf{R}_f(\mathbf{P}^T\mathbf{Q}_f\mathbf{R}_f)^{-1}\mathbf{P}^T \\ &= \mathbf{Q}_f(\mathbf{P}^T\mathbf{Q}_f)^{-1}\mathbf{P}^T. \end{aligned}$$

It does, however, provide an orthonormal basis to which the usual DEIM error estimate can be applied without introducing an additional dependence on the conditioning of the particular basis matrix $\mathbf{M}\Phi_e$.

Finally, if $\mathbf{M} = \mathbf{I}$, then the SNS construction reduces to

$$\Phi_f = \Phi_e.$$

In the minimal case $\Phi_e = \Phi$, the same POD basis is therefore used both to represent the state and to construct the nonlinear approximation. This special case makes the principal benefit of SNS particularly transparent: solution snapshots are reused, and the separate nonlinear snapshot collection and its SVD are removed from the offline workflow.

Chapter 2

Numerical experiments

2.1 Two-dimensional nonlinear function approximation

We begin the numerical experiments with a direct application of DEIM to a parameter-dependent family of scalar functions. This example does not come from a differential equation. Its purpose is to show the interpolation mechanism in a setting where the selected spatial points can be visualized directly. Let

$$\mathcal{D} = [0, 1]^2, \quad \mathcal{P} = [0.2, 0.8]^2.$$

For $\mu = (\mu_1, \mu_2) \in \mathcal{P}$, consider the function

$$f(x, y; \mu_1, \mu_2) = \exp\left(-40\left((x - \mu_1)^2 + (y - \mu_2)^2\right)\right), \quad (x, y) \in \mathcal{D}. \quad (2.1)$$

Thus the parameters determine the centre of a localized bump inside the square. The spatial domain is discretized by an 80×80 uniform grid, so each function value is represented by a vector in $\mathbb{R}^{6400}$. The training set is a 15×15 uniform grid in the parameter rectangle $\mathcal{P}$, giving 225 snapshots. The snapshot matrix is

$$F = \begin{bmatrix} f(\mu^{(1)}) & f(\mu^{(2)}) & \cdots & f(\mu^{(225)}) \end{bmatrix} \in \mathbb{R}^{6400 \times 225},$$

where each column is the vector of values of (2.1) on the spatial grid. We compute the SVD of F and take the first m left singular vectors as the basis $\mathbf{U}_m \in \mathbb{R}^{6400 \times m}$.

We compare two choices of interpolation indices: the standard greedy DEIM selection and QDEIM. Once the indices have been chosen, both methods use the same interpolation formula

$$f(\mu) \approx \mathbf{U}_m(P^T \mathbf{U}_m)^{-1} P^T f(\mu).$$

Figure 2.1 shows the interpolation points selected by DEIM and QDEIM for $m = 40$. Since the parameters move the centre of the bump inside $[0.2, 0.8]^2$, the selected points are concentrated mostly in the interior part of the square.

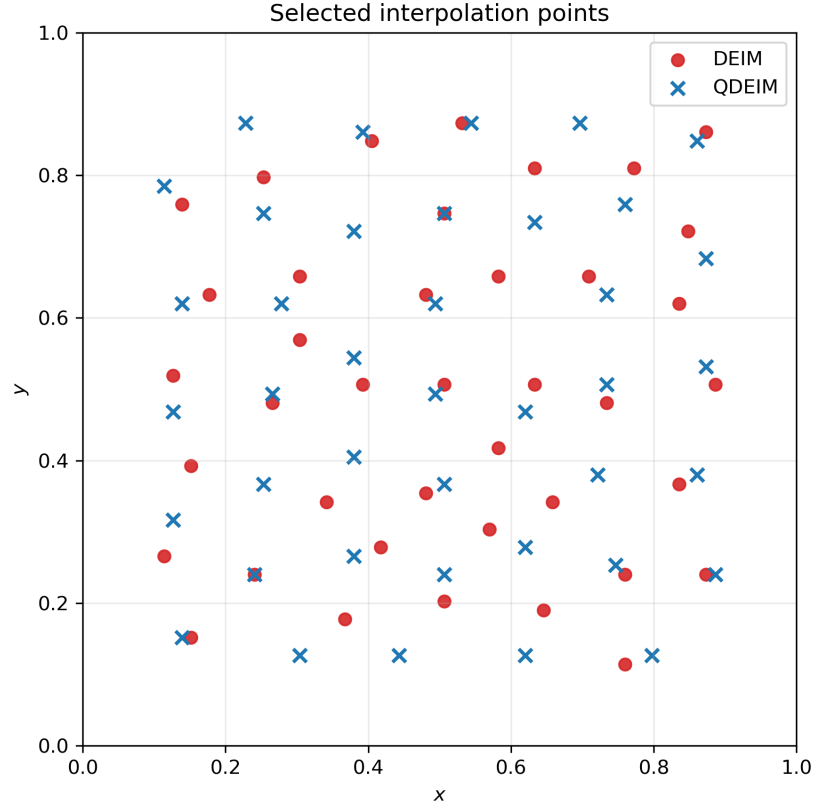


Figure 2.1: Spatial interpolation points selected by DEIM and QDEIM for the two-dimensional function family, using $m = 40$ basis vectors.

For a representative unseen parameter value $\mu = (0.63, 0.41)$, Figure 2.2 compares the exact function with the DEIM and QDEIM reconstructions. Both methods reproduce the location and shape of the localized bump.

Finally, Figure 2.3 compares the mean relative reconstruction error for $m = 5, \dots, 100$ on four unseen parameter values,

$$(0.25, 0.35), \quad (0.37, 0.71), \quad (0.52, 0.48), \quad (0.74, 0.62).$$

If the test parameters are denoted by $\mu_{\text{test}}^{(1)}, \dots, \mu_{\text{test}}^{(N_{\text{test}})}$, then for each value of m we compute

$$E_m = \frac{1}{N_{\text{test}}} \sum_{j=1}^{N_{\text{test}}} \frac{\|f(\mu_{\text{test}}^{(j)}) - \tilde{f}_m(\mu_{\text{test}}^{(j)})\|_2}{\|f(\mu_{\text{test}}^{(j)})\|_2},$$

where $\tilde{f}_m$ denotes either the DEIM or QDEIM reconstruction with m interpolation points. As expected, the error decreases as the number of retained basis vectors increases. In this

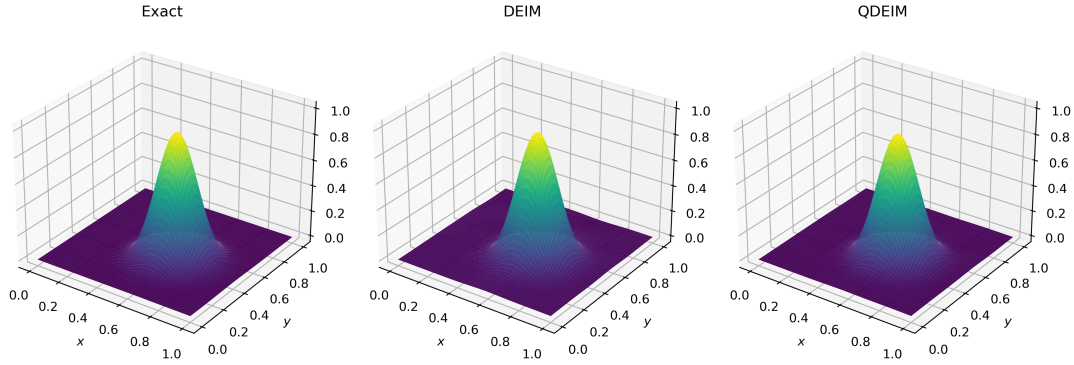


Figure 2.2: Exact function, DEIM reconstruction, and QDEIM reconstruction for the unseen parameter value $\mu = (0.63, 0.41)$, shown as surfaces over the spatial domain.

example DEIM and QDEIM have comparable accuracy, while selecting somewhat different interpolation points.

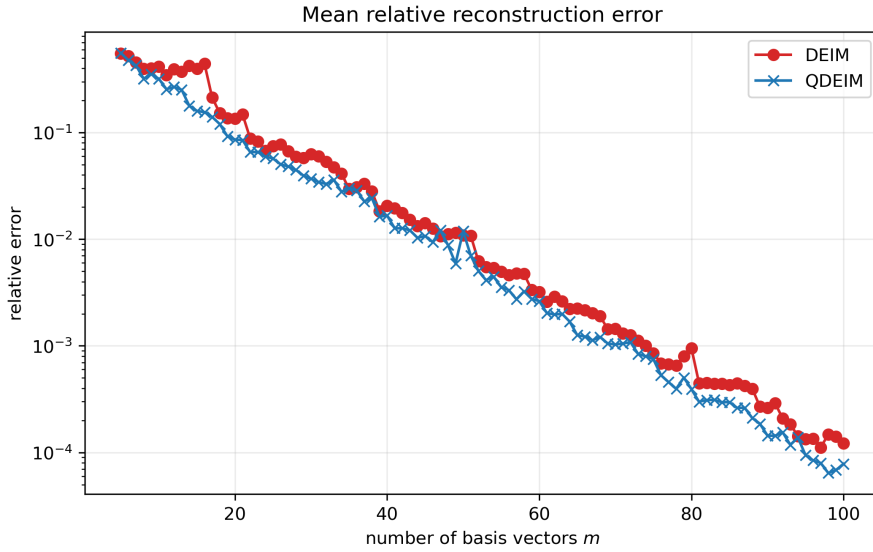


Figure 2.3: Mean relative reconstruction error for DEIM and QDEIM on unseen parameter values, for $m = 5, \dots, 100$.

2.1.1 Restriction to an admissible part of the domain

We now use the same snapshot matrix to illustrate the restricted selection from Section 1.8. Suppose that interpolation points cannot be placed in the central square

$$\mathcal{D}_{\text{forbidden}} = [0.35, 0.65] \times [0.35, 0.65].$$

The admissible index set consists of all grid points outside this square. Restricted QDEIM is obtained by applying column-pivoted QR only to the columns of $\mathbf{U}_m^T$ associated with these admissible grid points.

Figure 2.4 compares the unrestricted and restricted selections for $m = 40$. Several points preferred by unrestricted QDEIM lie inside the inaccessible square. Restricted QDEIM replaces them by points outside the square while retaining a full-rank interpolation matrix.

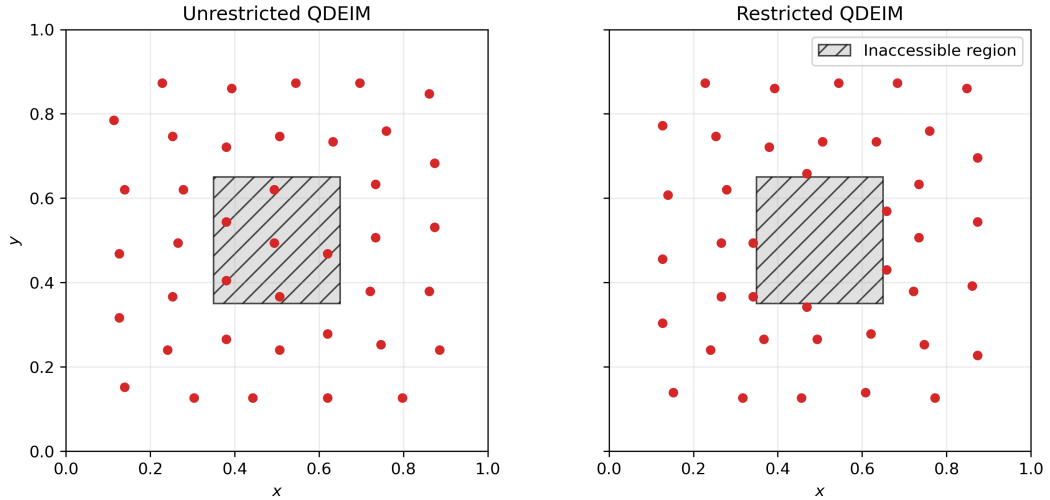


Figure 2.4: Unrestricted and restricted QDEIM interpolation points for $m = 40$. The shaded central square is inaccessible to the restricted selection.

The rank condition (1.14) is satisfied for every tested dimension $m = 5, 10, \dots, 80$. Nevertheless, full rank does not imply that the restricted interpolation problem is as well conditioned as the unrestricted one. At $m = 40$, the interpolation constants are

$$\|(\mathbf{P}_Q^T \mathbf{U}_{40})^{-1}\|_2 \approx 31.1, \quad \|(\mathbf{P}_{\mathcal{A}}^T \mathbf{U}_{40})^{-1}\|_2 \approx 196.2,$$

where $\mathbf{P}_Q$ denotes unrestricted QDEIM and $\mathbf{P}_{\mathcal{A}}$ denotes restricted QDEIM. The corresponding mean relative errors on the four test parameters are

$$E_{40}^{\text{QDEIM}} \approx 1.67 \cdot 10^{-2}, \quad E_{40}^{\text{restricted}} \approx 6.20 \cdot 10^{-2}.$$

The restriction therefore produces a noticeable, but still moderate, deterioration in this example.

Figure 2.5 shows the same comparison over all tested dimensions. The restriction does not alter the POD basis or its projection error. It reduces the available choices for the interpolation matrix and can therefore increase the interpolation constant and amplify the basis residual, exactly as predicted by (1.18). The experiment also illustrates why the rank test alone is insufficient: the admissible basis has full column rank, but the larger interpolation constant shows that its directions are not represented equally well outside the central square.

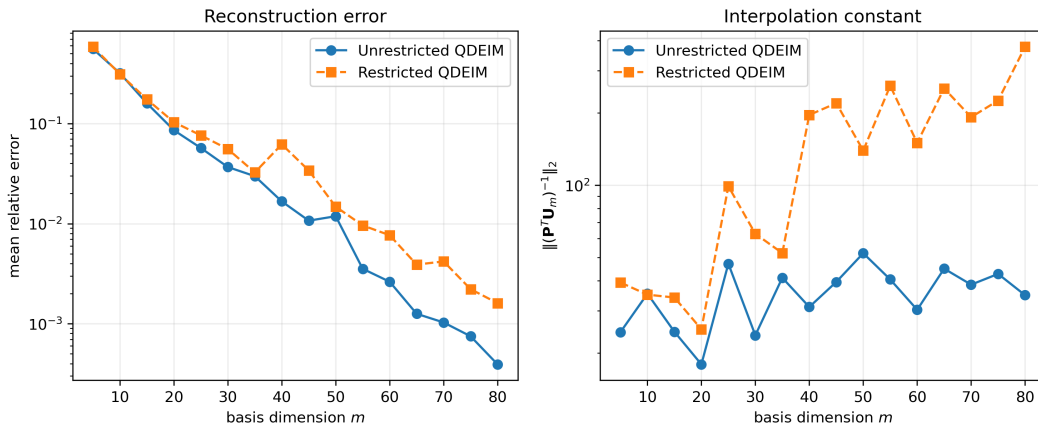


Figure 2.5: Mean reconstruction error and interpolation constant for unrestricted and restricted QDEIM.

2.2 The FitzHugh–Nagumo system

The FitzHugh–Nagumo system is a two-variable simplification of the Hodgkin–Huxley model of neuronal activity. It is a standard nonlinear model-reduction benchmark because a low-dimensional approximation must retain the repeated excitation and recovery cycles of the solution [5].

2.2.1 Full-order model and training simulation

We consider

$$\varepsilon v_t = \varepsilon^2 v_{xx} + f(v) - w + c, \quad (2.2)$$

$$w_t = bv - \gamma w + c, \quad (2.3)$$

on $[0, L] \times [0, T]$, where

$$f(v) = v(v - 0.1)(1 - v).$$

The parameters and data are

$$L = 1, \quad \varepsilon = 0.015, \quad b = 0.5, \quad \gamma = 2, \quad c = 0.05,$$

$$v(x, 0) = w(x, 0) = 0, \quad v_x(0, t) = -i_0(t), \quad v_x(L, t) = 0,$$

with boundary stimulus

$$i_0(t) = 5 \cdot 10^4 t^3 e^{-15t}.$$

Let

$$x_i = (i - 1)h, \quad i = 1, \dots, N, \quad h = \frac{L}{N - 1},$$

so that $x_1 = 0$ and $x_N = L$. At the interior nodes, the usual centered difference approximation gives

$$v_{xx}(x_i, t) \approx \frac{v_{i+1} - 2v_i + v_{i-1}}{h^2}, \quad i = 2, \dots, N - 1.$$

For completeness, we state precisely how the boundary rows used in the implementation are obtained. Introduce ghost values v_0 and v_{N+1} and enforce the Neumann conditions by

$$\frac{v_1 - v_0}{h} = -i_0(t), \quad \frac{v_{N+1} - v_N}{h} = 0.$$

Hence

$$v_0 = v_1 + hi_0(t), \quad v_{N+1} = v_N.$$

Substitution into the second differences at the two boundary nodes yields

$$v_{xx}(x_1, t) \approx \frac{v_2 - v_1}{h^2} + \frac{i_0(t)}{h}, \quad v_{xx}(x_N, t) \approx \frac{v_{N-1} - v_N}{h^2}.$$

Therefore, with

$$D = \frac{1}{h^2} \begin{bmatrix} -1 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \\ & & & 1 & -1 \end{bmatrix} \in \mathbb{R}^{N \times N},$$

the vectors

$$v(t) = [v_1(t) \ \cdots \ v_N(t)]^T, \quad w(t) = [w_1(t) \ \cdots \ w_N(t)]^T$$

satisfy

$$\begin{aligned} \dot{v} &= \varepsilon D v + \frac{1}{\varepsilon} f(v) - \frac{1}{\varepsilon} w + \frac{c}{\varepsilon} \mathbf{1} + \frac{\varepsilon}{h} e_1 i_0(t), \\ \dot{w} &= b v - \gamma w + c \mathbf{1}, \end{aligned}$$

where f is applied componentwise.

Define the full state $y = [v^T \ w^T]^T \in \mathbb{R}^{2N}$. The semidiscrete system can then be written as

$$\dot{y}(t) = \mathbf{A} y(t) + \mathcal{G}(y(t)) + r + d i_0(t), \quad y(0) = 0, \quad (2.4)$$

where

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} \varepsilon D & -\varepsilon^{-1} \mathbf{I} \\ b \mathbf{I} & -\gamma \mathbf{I} \end{bmatrix}, \quad \mathcal{G}(y) = \begin{bmatrix} \varepsilon^{-1} f(v) \\ 0 \end{bmatrix}, \\ r &= \begin{bmatrix} c \varepsilon^{-1} \mathbf{1} \\ c \mathbf{1} \end{bmatrix}, \quad d = \begin{bmatrix} \varepsilon h^{-1} e_1 \\ 0 \end{bmatrix}. \end{aligned}$$

This decomposition separates the cubic nonlinear term, which will be approximated by DEIM, from the linear, constant, and known time-dependent terms, which can be projected once and evaluated directly in reduced coordinates.

The full dimension is $2N = 1024$.

2.2.2 POD–DEIM construction

The full system is integrated first, and its values are saved at $t_1, \dots, t_{N_t}$. From the same trajectory we form the state and nonlinear snapshot matrices

$$\mathbf{Y} = [y(t_1) \ \cdots \ y(t_{N_t})] \in \mathbb{R}^{2N \times N_t}, \quad (2.5)$$

and

$$\mathbf{G} = [\mathcal{G}(y(t_1)) \ \cdots \ \mathcal{G}(y(t_{N_t}))] \in \mathbb{R}^{2N \times N_t}. \quad (2.6)$$

The trajectory begins at the origin, and in this experiment no affine shift or mean subtraction is used. Let

$$\mathbf{Y} = \mathbf{V}\Sigma_Y\mathbf{W}^T, \quad \mathbf{G} = \mathbf{U}\Sigma_G\mathbf{Z}^T$$

be reduced singular value decompositions. The first k and m left singular vectors define

$$\mathbf{V}_k = \mathbf{V}(:, 1 : k), \quad \mathbf{U}_m = \mathbf{U}(:, 1 : m).$$

Thus, the construction uses the usual two-snapshot-matrix POD–DEIM workflow: $\mathbf{V}_k$ represents the state, whereas $\mathbf{U}_m$ represents only the nonlinear mapping $\mathcal{G}$.

We take $k = m = 10$ so that the state and nonlinear dimensions can be compared at the same truncation level.

2.2.3 Reduced POD–DEIM model

With

$$y(t) \approx \mathbf{V}_k \widehat{y}(t),$$

the Galerkin projection of (2.4) would still require the full vector $\mathcal{G}(\mathbf{V}_k \widehat{y})$. Let

$$\mathbf{P} = \begin{bmatrix} e_{p_1} & \cdots & e_{p_m} \end{bmatrix}$$

be generated by the original DEIM selection applied to $\mathbf{U}_m$. Using the DEIM approximation and collecting the quantities that can be precomputed gives the reduced system

$$\begin{aligned} \dot{\widehat{y}} = & \underbrace{\mathbf{V}_k^T \mathbf{A} \mathbf{V}_k}_{\mathbf{A}_r} \widehat{y} + \underbrace{\mathbf{V}_k^T \mathbf{U}_m (\mathbf{P}^T \mathbf{U}_m)^{-1} \mathbf{P}^T}_{\mathbf{B}_{\text{DEIM}}} \mathcal{G}(\mathbf{V}_k \widehat{y}) \\ & + \mathbf{V}_k^T r + \mathbf{V}_k^T d i_0(t). \end{aligned} \quad (2.7)$$

Equation (2.7) is the reduced model integrated in the numerical experiment. The inverse displayed in (2.7) is not formed in the implementation; the corresponding linear system is solved during the offline construction of $\mathbf{B}_{\text{DEIM}}$.

In this example the nonlinear mapping is pointwise and its recovery block is identically zero. Consequently, all selected indices associated with the nonzero nonlinear basis lie in the voltage block. If $\mathbf{V}_k^{(v)} \in \mathbb{R}^{N \times k}$ denotes the first N rows of $\mathbf{V}_k$ and $\mathbf{P}_v \in \mathbb{R}^{N \times m}$ selects the corresponding voltage coordinates, then

$$\mathbf{P}^T \mathcal{G}(\mathbf{V}_k \widehat{y}) = \frac{1}{\varepsilon} f(\mathbf{P}_v^T \mathbf{V}_k^{(v)} \widehat{y}).$$

Only m scalar evaluations of f are therefore needed online. The next experiment shows why this particularly convenient commutation property must be replaced by a larger dependency stencil for a locally coupled nonlinear mapping.

Both the full and reduced systems are integrated on $[0, 8]$ in Python using the SciPy function `solve_ivp` with the variable-step BDF method, and their values are stored at 300 equally spaced output times. The solver defaults $\text{rtol} = 10^{-3}$ and $\text{atol} = 10^{-6}$ are used.

2.2.4 Numerical results

Figure 2.6 shows the normalized singular values of the two snapshot matrices. To quantify their truncation directly, define

$$\eta_Y(k) = \frac{\|(\mathbf{I} - \mathbf{V}_k \mathbf{V}_k^T) \mathbf{Y}\|_F}{\|\mathbf{Y}\|_F}, \quad \eta_G(m) = \frac{\|(\mathbf{I} - \mathbf{U}_m \mathbf{U}_m^T) \mathbf{G}\|_F}{\|\mathbf{G}\|_F}.$$

At $k = m = 10$, these residuals are

$$\eta_Y(10) = 2.79 \cdot 10^{-4}, \quad \eta_G(10) = 4.25 \cdot 10^{-5}.$$

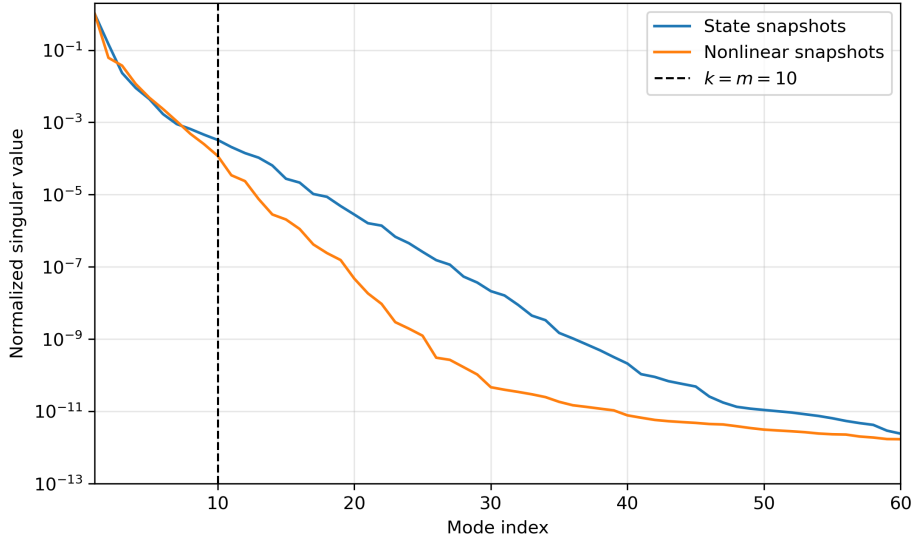


Figure 2.6: Normalized singular values of the state and nonlinear snapshot matrices. The dashed line marks the retained dimensions $k = m = 10$.

The interpolation matrix at this dimension satisfies

$$\text{cond}_2(\mathbf{P}^T \mathbf{U}_{10}) = 13.60, \quad \|(\mathbf{P}^T \mathbf{U}_{10})^{-1}\|_2 = 15.50.$$

Thus neither the POD truncation nor the selected interpolation system is severely ill-conditioned. Nevertheless, the error of an integrated reduced trajectory is not the same quantity as the projection residual of the training snapshots. With

$$\widetilde{\mathbf{Y}} = [\mathbf{V}_k \widehat{\mathbf{y}}(t_1) \quad \cdots \quad \mathbf{V}_k \widehat{\mathbf{y}}(t_{N_t})],$$

we obtain

$$\frac{\|\mathbf{Y} - \widetilde{\mathbf{Y}}\|_F}{\|\mathbf{Y}\|_F} = 5.59 \cdot 10^{-2}. \quad (2.8)$$

The corresponding block errors are $5.71 \cdot 10^{-2}$ for v and $1.87 \cdot 10^{-2}$ for w .

Figure 2.7 compares the full and reduced values at the central spatial node and also reports the time-dependent full-state error

$$e_j = \frac{\|y(t_j) - \mathbf{V}_k \widehat{y}(t_j)\|_2}{\|y(t_j)\|_2},$$

with $e_1 = 0$ at the zero initial condition. The two local traces remain close over the four excitation cycles. The full-state error, however, develops short peaks near the rapid transitions. Its largest value is 0.922 at $t \approx 7.572$. At that instant $\|y(t)\|_2 \approx 4.02$, so the peak is not an artifact of division by a nearly zero reference norm. Inspection of the neighbouring saved states shows that the reduced solution slightly lags behind the full solution during this rapid transition. Comparing the reduced state at $t \approx 7.572$ with the full state one output step earlier, a time difference of approximately 0.0268, reduces the relative discrepancy from 0.922 to 0.209. Hence, much of the peak is caused by a small temporal displacement of the transition.

This experiment illustrates a distinction that is easy to miss when only snapshot projection errors are inspected. Rapid singular-value decay shows that the sampled vectors are well represented by low-dimensional spaces, but the reduced differential equation is then integrated over time. Projection and interpolation errors can perturb the phase of a sharp nonlinear transition and accumulate along the trajectory. The quality of a dynamical reduced model must therefore be assessed on its integrated solution, not only on the singular values of its training snapshots.

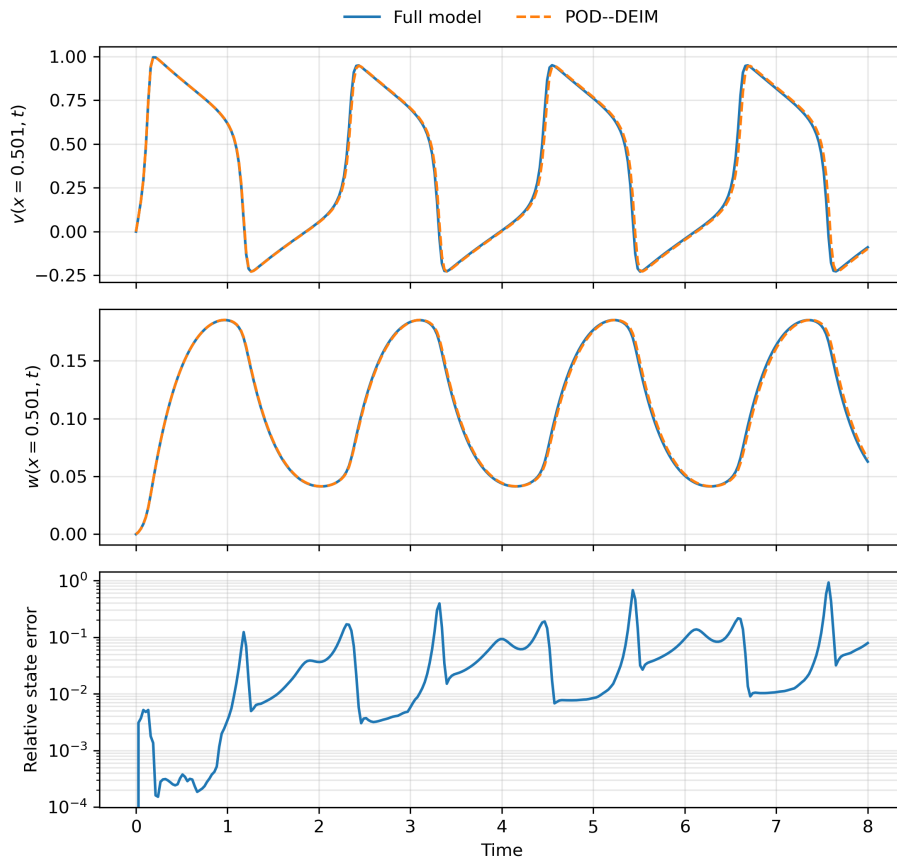


Figure 2.7: Full and POD-DEIM trajectories at the central spatial node (top and middle) and the relative full-state error at the saved times (bottom), for $k = m = 10$.

2.3 Nonlinear RC-ladder circuit

The previous numerical examples have a particularly convenient property. If the nonlinear mapping is pointwise,

$$F(v) = \begin{bmatrix} f(v_1) & \cdots & f(v_n) \end{bmatrix}^T,$$

and $\mathbf{P}$ selects m coordinates, then evaluation and coordinate selection commute:

$$\mathbf{P}^T F(v) = f(\mathbf{P}^T v), \quad (2.9)$$

where the scalar function f is applied componentwise on the right-hand side. Consequently, after introducing the reduced approximation $v \approx \mathbf{V}_r \widehat{v}$, only the selected state entries $\mathbf{P}^T \mathbf{V}_r \widehat{v}$ are needed.

Identity (2.9) is useful, but it is not part of the general DEIM theory. DEIM requires the selected values $\mathbf{P}^T F(v)$ to be available at low cost; it does not require a componentwise nonlinear mapping. The nonlinear RC-ladder circuit provides a simple example for which (2.9) does not hold. Each component of its vector field depends on voltage differences involving neighbouring circuit nodes. We must therefore distinguish the selected output coordinates from the state coordinates required to evaluate them.

2.3.1 Full-order model and training simulation

We use the nonlinear RC-ladder benchmark described by Chen [6] and subsequently used in model-reduction studies such as [2, 9]. A reproducible description and a procedural implementation are also available from the MORwiki benchmark collection [12].

Let

$$x(t) = \begin{bmatrix} x_1(t) & \cdots & x_n(t) \end{bmatrix}^T \in \mathbb{R}^n$$

contain the voltages at the capacitor nodes. All capacitances are equal to one. Each nonlinear resistor consists of a unit resistor connected in parallel with a diode whose current–voltage characteristic is

$$g_D(z) = e^{40z} - 1.$$

The total current through the parallel connection is consequently

$$g(z) = g_D(z) + z = e^{40z} + z - 1. \quad (2.10)$$

An external current source $u(t)$ enters at the first node. Kirchhoff's current law gives

$$\dot{x}_1 = -g(x_1) - g(x_1 - x_2) + u(t), \quad (2.11)$$

$$\dot{x}_i = g(x_{i-1} - x_i) - g(x_i - x_{i+1}), \quad i = 2, \dots, n-1, \quad (2.12)$$

$$\dot{x}_n = g(x_{n-1} - x_n). \quad (2.13)$$

The local structure becomes especially clear in matrix form. Define the edge-voltage matrix

$$\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & -1 & 0 & \cdots & 0 \\ 0 & 1 & -1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 & -1 \end{bmatrix} \in \mathbb{R}^{n \times n}.$$

Then

$$\mathbf{C}x = \begin{bmatrix} x_1 & x_1 - x_2 & \cdots & x_{n-1} - x_n \end{bmatrix}^T.$$

If g is applied componentwise, equations (2.11)–(2.13) can be written as

$$\dot{x}(t) = \mathcal{F}(x(t), t) := -\mathbf{C}^T g(\mathbf{C}x(t)) + \mathbf{b}u(t), \quad \mathbf{b} = e_1. \quad (2.14)$$

The experimental parameters follow the RC-ladder example in [9] and are summarized in Table 2.1. The initial state is zero. Both the full and reduced systems are integrated in Python using the SciPy function `solve_ivp` with the variable-step BDF method.

Table 2.1: Setup of the nonlinear RC-ladder experiment.

Quantity	Value
Number of circuit nodes	$n = 1000$
Input current	$u(t) = e^{-t}$
Time interval	$[0, 7]$
Initial state	$x(0) = 0$
Number of saved snapshots	$N_t = 1425$
Time integrator	BDF
Relative and absolute solver tolerances	10^{-8} and 10^{-10}
Dimensions	$r = m = 4, 6, \dots, 20$

2.3.2 POD–DEIM and SNS construction

The solution snapshots and right-hand-side snapshots are

$$\mathbf{X} = \begin{bmatrix} x(t_1) & \cdots & x(t_{N_t}) \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} \mathcal{F}(x(t_1), t_1) & \cdots & \mathcal{F}(x(t_{N_t}), t_{N_t}) \end{bmatrix}.$$

Let $\mathbf{V}_r$ contain the first r left singular vectors of $\mathbf{X}$ and let $\mathbf{U}_m$ contain the first m left singular vectors of $\mathbf{R}$. We approximate the complete right-hand side, including the inexpensive input term, in every hyper-reduced model. This keeps the object being approximated identical when the conventional and SNS basis constructions are compared. The same snapshots and state basis are used by every reduced model.

The mass matrix is the identity, so the SNS nonlinear basis is obtained directly from the solution basis.

For standard DEIM and QDEIM, the right-hand-side basis is $\mathbf{U}_m$. For the SNS variants, $\mathbf{M} = \mathbf{I}$ and we take $m = r$ and use $\mathbf{V}_r$ itself as the right-hand-side basis. We apply both the original DEIM selection and QDEIM selection to each basis. This gives four hyper-reduced variants: DEIM and QDEIM use the right-hand-side snapshot basis $\mathbf{U}_m$, whereas SNS-DEIM and SNS-QDEIM use the solution basis $\mathbf{V}_r$. The comparison therefore separates the effect of the basis construction from the effect of the index-selection rule.

2.3.3 Exact evaluation on a local sampling stencil

Let

$$\mathbf{P} = \begin{bmatrix} e_{p_1} & \cdots & e_{p_m} \end{bmatrix} \in \mathbb{R}^{n \times m}$$

select the output coordinates of $\mathcal{F}$. Equation (2.12) shows that evaluating $\mathcal{F}_{p_j}(x, t)$ generally requires $x_{p_{j-1}}$, x_{p_j} , and $x_{p_{j+1}}$. Hence

$$\mathbf{P}^T \mathcal{F}(x, t) \neq \mathcal{F}(\mathbf{P}^T x, t)$$

in general. In fact, the mapping on the right would have a different input dimension and would no longer describe the original circuit.

To obtain the selected values without forming the complete vector field, denote the selected node set by

$$\mathcal{I} = \{p_1, \dots, p_m\},$$

and let $\mathcal{E}(\mathcal{I})$ contain all circuit edges incident to these nodes. If $\ell = |\mathcal{E}(\mathcal{I})|$, define

$$\mathbf{E} = \begin{bmatrix} e_{q_1} & \cdots & e_{q_\ell} \end{bmatrix} \in \mathbb{R}^{n \times \ell}$$

to select those edges. The rows of $\mathbf{C}\mathbf{P}$ associated with edges outside $\mathcal{E}(\mathcal{I})$ are zero, and therefore

$$\mathbf{P}^T \mathbf{C}^T = \mathbf{P}^T \mathbf{C}^T \mathbf{E} \mathbf{E}^T.$$

It follows exactly that

$$\mathbf{P}^T \mathcal{F}(x, t) = -\mathbf{P}^T \mathbf{C}^T \mathbf{E} g(\mathbf{E}^T \mathbf{C} x) + \mathbf{P}^T \mathbf{b} u(t). \quad (2.15)$$

Now introduce the POD approximation

$$x(t) \approx \mathbf{V}_r \widehat{x}(t), \quad \mathbf{V}_r \in \mathbb{R}^{n \times r}, \quad \mathbf{V}_r^T \mathbf{V}_r = \mathbf{I}_r,$$

and precompute

$$\mathbf{H} = \mathbf{E}^T \mathbf{C} \mathbf{V}_r \in \mathbb{R}^{\ell \times r}, \quad \mathbf{L} = -\mathbf{P}^T \mathbf{C}^T \mathbf{E} \in \mathbb{R}^{m \times \ell}. \quad (2.16)$$

Equation (2.15) becomes

$$\boxed{\mathbf{P}^T \mathcal{F}(\mathbf{V}_r \widehat{x}, t) = \mathbf{L}g(\mathbf{H}\widehat{x}) + \mathbf{P}^T \mathbf{b} u(t).} \quad (2.17)$$

Thus the online evaluation consists of a multiplication by $\mathbf{H}$, ℓ scalar evaluations of g , and a multiplication by $\mathbf{L}$. A node is incident to at most two edges, so $\ell \leq 2m$. Equivalently, the required state-node set

$$\mathcal{J} = \bigcup_{p \in \mathcal{I}} \{p-1, p, p+1\} \cap \{1, \dots, n\}$$

contains at most $3m$ nodes. The cost of (2.17) is therefore independent of the full-order dimension n when m and r are fixed.

2.3.4 Reduced models

The POD–Galerkin model is

$$\dot{\widehat{x}} = \mathbf{V}_r^T \mathcal{F}(\mathbf{V}_r \widehat{x}, t). \quad (2.18)$$

Once either index-selection rule has produced $\mathbf{P}$, the reduced equation is

$$\dot{\widehat{x}} = \mathbf{V}_r^T \mathbf{U}_m (\mathbf{P}^T \mathbf{U}_m)^{-1} [\mathbf{L}g(\mathbf{H}\widehat{x}) + \mathbf{P}^T \mathbf{b} u(t)]. \quad (2.19)$$

For the SNS variants, Equation (2.19) simplifies to

$$\dot{\widehat{x}} = (\mathbf{P}^T \mathbf{V}_r)^{-1} [\mathbf{L}g(\mathbf{H}\widehat{x}) + \mathbf{P}^T \mathbf{b} u(t)]. \quad (2.20)$$

Equations (2.18), (2.19), and (2.20) are the reduced models integrated in the numerical comparison. All of them use the reduced initial condition $\widehat{x}(0) = \mathbf{V}_r^T x(0) = 0$.

2.3.5 Numerical results

Figure 2.8 shows that both snapshot matrices have a rapid singular-value decay. The right-hand-side singular values decay more slowly than the solution singular values over the first modes, which explains why a basis constructed directly from $\mathbf{R}$ and one constructed from $\mathbf{X}$ need not behave identically at the same dimension.

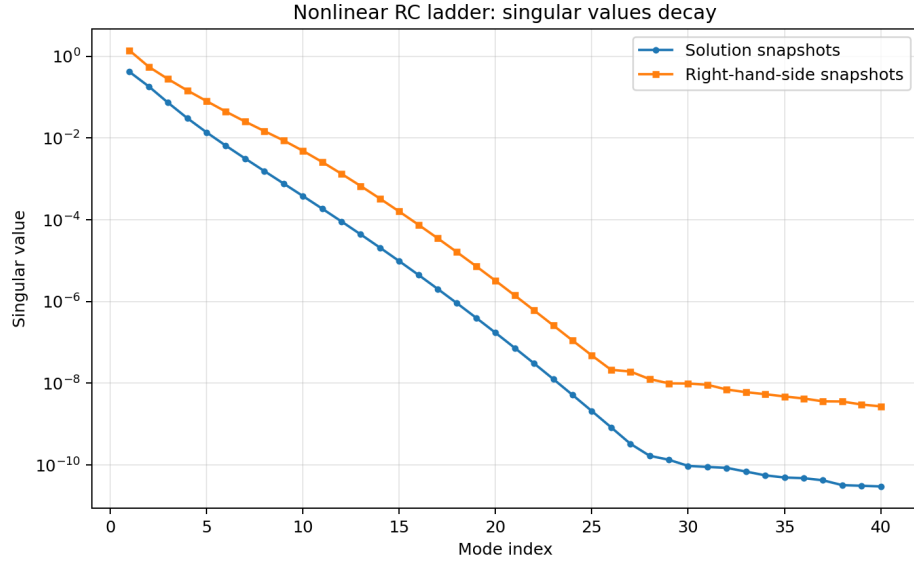


Figure 2.8: Singular values of the solution and right-hand-side snapshot matrices for the nonlinear RC ladder.

For every interpolation matrix and every dimension, the implementation compares the local expression (2.17) with direct sampling of the complete vector field. At $r = m = 10$, the largest relative discrepancy among the four variants is $5.2 \cdot 10^{-16}$. Thus, the use of the local stencil introduces no additional approximation; all reported differences arise from projection, basis truncation, and index selection.

Figure 2.9 displays the selected output nodes and their required state-node closures for $r = m = 10$. Although each method selects ten outputs, DEIM and QDEIM require only 21 state nodes, SNS-DEIM requires 22, and SNS-QDEIM requires 24. These are small subsets of the complete 1000-node circuit and confirm that the sampled vector field can be evaluated without forming the full state or the full right-hand side.

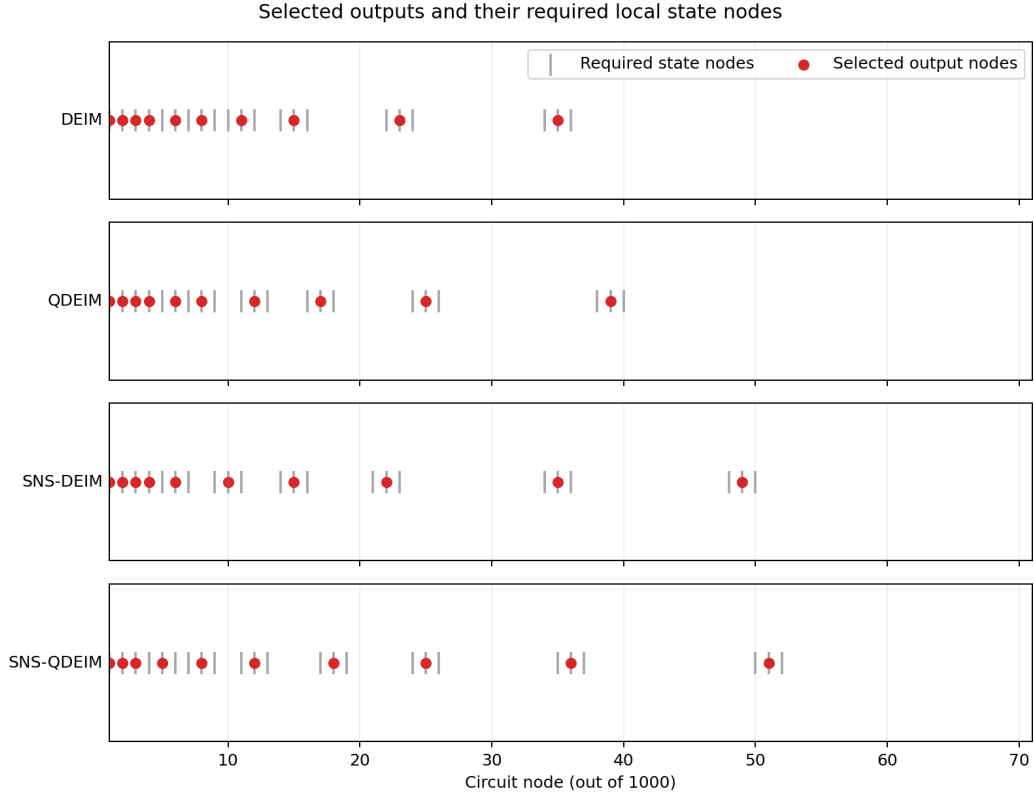


Figure 2.9: Output nodes selected at $r = m = 10$ and the local state nodes required to evaluate the corresponding vector-field entries exactly.

We measure the full-trajectory error by

$$\mathcal{E}_{\text{traj}} = \frac{\|\mathbf{X} - \tilde{\mathbf{X}}\|_F}{\|\mathbf{X}\|_F}, \quad (2.21)$$

where $\tilde{\mathbf{X}}$ contains the reconstructed reduced trajectory. The results in Figure 2.10 show convergence for all five reduced models. At $r = m = 10$, the errors for POD, DEIM, QDEIM, SNS-DEIM, and SNS-QDEIM are, respectively,

$$5.68 \cdot 10^{-4}, \quad 4.90 \cdot 10^{-3}, \quad 3.25 \cdot 10^{-3}, \quad 1.49 \cdot 10^{-3}, \quad 6.40 \cdot 10^{-4}.$$

At dimension 20 all four hyper-reduced errors are below 10^{-6} . In this experiment SNS-QDEIM stays close to the POD error, while QDEIM generally improves upon the standard right-hand-side DEIM model once the dimension is moderate. This is an observation for this trajectory, not a general ordering of the methods.

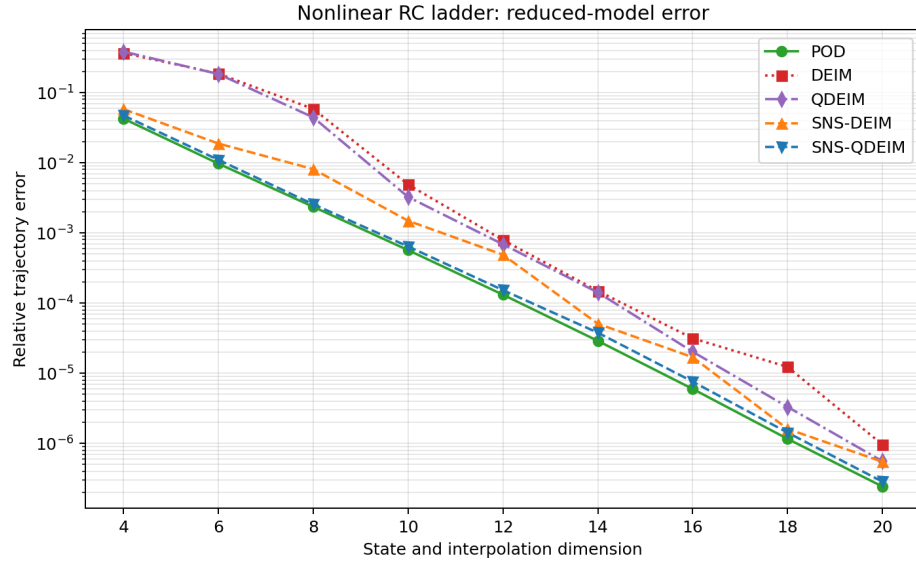


Figure 2.10: Relative full-trajectory error as the state and interpolation dimensions are increased together.

Finally, Figure 2.11 shows the voltage at the first node, which is the quantity of interest used in [9]. The curves are visually indistinguishable at $r = m = 10$, so the lower panel shows their absolute errors. The relative node-1 errors range from $1.14 \cdot 10^{-5}$ for POD to $5.34 \cdot 10^{-5}$ for SNS-QDEIM. The ordering differs from the full-trajectory metric: a reduced model can be more accurate over the complete state while being slightly less accurate in one selected quantity of interest. Both measures are therefore reported.

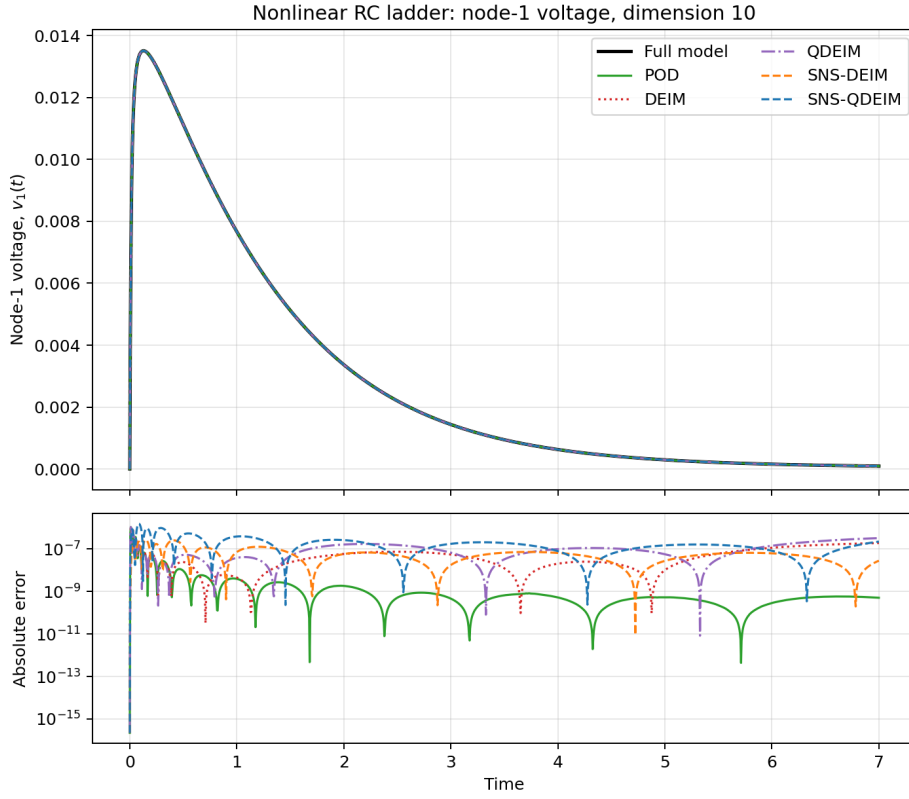


Figure 2.11: Voltage at the first circuit node and its absolute reconstruction error for $r = m = 10$.

The experiment demonstrates the practical condition underlying the claimed online efficiency of DEIM-type methods. Pointwise commutation as in (2.9) is sufficient but not necessary. For a locally coupled nonlinear mapping, an exact dependency closure such as (2.17) provides the required selected values at a cost independent of the full-order dimension.

2.4 Sparse reconstruction of a parametric cylinder wake

The preceding examples use DEIM to approximate a nonlinear mapping inside a reduced differential equation. In this experiment we use the same interpolation operator in a different, but mathematically natural, role. We assume that a time-dependent vorticity field is known only at a small number of spatial points and reconstruct the complete field from those values. Thus, this section concerns sparse state reconstruction; it does not construct or integrate a hyper-reduced Navier–Stokes model.

2.4.1 Full-order problem and training simulation

The full-order dataset generator is adapted from the cylinder-wake example in the FEniCS hands-on tutorial [3]. The tutorial uses the legacy FEniCS interface; here the solver is reimplemented in Python with FEniCSx/DOLFINx and extended to generate a parametric vorticity dataset. Gmsh is used to generate the mesh, and the algebraic systems are solved through PETSc using LU factorization, with MUMPS when available. We retain the channel geometry, viscosity, parabolic inlet profile, boundary conditions, Taylor–Hood elements, and Crank–Nicolson time discretization. The parameter-dependent inlet amplitude, smooth startup ramp, symmetry-breaking perturbations, Picard linearization, smaller time step, and vorticity-snapshot output are additions made for the present experiment.

Let

$$\Omega = ((0, 2.2) \times (0, 0.41)) \setminus \overline{B_{0.05}((0.2, 0.2))}$$

be a two-dimensional channel containing a circular obstacle. For the parameter

$$\mu = U_{\max} \in [0.7, 1.0],$$

we solve the incompressible Navier–Stokes equations

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u - \nu \Delta u + \nabla p = 0 \quad \text{in } \Omega \times (0, T], \quad (2.22)$$

$$\nabla \cdot u = 0 \quad \text{in } \Omega \times (0, T], \quad (2.23)$$

with kinematic viscosity $\nu = 10^{-3}$ and final time $T = 8$. Here $u = (u_1, u_2)^T$ is the velocity and p is the pressure. No-slip conditions are imposed on the channel walls and the cylinder, while a parabolic velocity profile is prescribed at the inlet. Its horizontal amplitude is ramped according to

$$a(t; \mu) = \mu \sin\left(\frac{\pi}{2} \min\left\{\frac{t}{4}, 1\right\}\right).$$

A small, time-decaying vertical inlet perturbation is added to initiate an asymmetric wake:

$$k(t; \mu) = \begin{cases} 0.02\mu \left(1 + \cos\left(\frac{\pi t}{6}\right)\right) \sin\left(\frac{2\pi t}{0.6}\right), & 0 \leq t \leq 6, \\ 0, & t > 6. \end{cases}$$

Consequently, with $H = 0.41$, the inlet condition is

$$u_{\text{in}}(y, t; \mu) = \begin{bmatrix} 4a(t; \mu)y(H - y)/H^2 \\ k(t; \mu) \sin(\pi y/H) \end{bmatrix}.$$

The initial pressure is zero, and the initial velocity contains the small vertical perturbation

$$u(x, y, 0; \mu) = \begin{bmatrix} 0 \\ 10^{-2} \sin(2\pi x/2.2) \sin(\pi y/H) \end{bmatrix}.$$

At the outflow we use the natural boundary condition produced by the weak formulation.

The spatial discretization uses Taylor–Hood finite elements. The velocity is approximated by continuous piecewise quadratic functions and the pressure by continuous piecewise linear functions. If

$$u^{j-\frac{1}{2}} = \frac{1}{2}(u^j + u^{j-1}),$$

then one time step of the Picard-linearized Crank–Nicolson scheme is: find (u^j, p^j) such that

$$\begin{aligned} \left(\frac{u^j - u^{j-1}}{\Delta t}, v \right) + \nu \left(\nabla u^{j-\frac{1}{2}}, \nabla v \right) + \left((u^{j-1} \cdot \nabla) u^{j-\frac{1}{2}}, v \right) \\ - \left(p^j, \nabla \cdot v \right) - \left(q, \nabla \cdot u^j \right) = 0 \end{aligned} \quad (2.24)$$

for every discrete test pair (v, q) . We use $\Delta t = 0.01$ and save every second time step.

The reduced quantity is the scalar vorticity

$$\omega(x, t; \mu) = \frac{\partial u_2}{\partial x}(x, t; \mu) - \frac{\partial u_1}{\partial y}(x, t; \mu). \quad (2.25)$$

It is interpolated into the continuous piecewise linear finite-element space. Let $n = 11994$ be the number of vorticity degrees of freedom and let $t_1, \dots, t_{N_t}$, with $N_t = 400$, be the saved times. For a fixed parameter μ_i , the vorticity snapshot matrix is

$$\mathbf{W}(\mu_i) = \begin{bmatrix} \omega(t_1; \mu_i) & \omega(t_2; \mu_i) & \cdots & \omega(t_{N_t}; \mu_i) \end{bmatrix} \in \mathbb{R}^{n \times N_t}, \quad (2.26)$$

where a field and its vector of nodal values are denoted by the same symbol.

2.4.2 POD, DEIM, and QDEIM construction

Suppose that the training parameters are $\mu_1, \dots, \mu_{N_\mu}$. Concatenating the corresponding trajectories gives

$$\mathbf{X} = \begin{bmatrix} \mathbf{W}(\mu_1) & \cdots & \mathbf{W}(\mu_{N_\mu}) \end{bmatrix} \in \mathbb{R}^{n \times N}, \quad N = N_\mu N_t. \quad (2.27)$$

We first subtract the mean training field

$$\bar{\omega} = \frac{1}{N} \mathbf{X} \mathbf{1}, \quad \mathbf{X}_c = \mathbf{X} - \bar{\omega} \mathbf{1}^T.$$

The ordering of times and parameters within (2.27) is immaterial to the POD construction. For any permutation matrix Π , set $\tilde{\mathbf{X}} = \mathbf{X} \Pi$. Because $\Pi \mathbf{1} = \mathbf{1}$ and $\Pi \Pi^T = \mathbf{I}$, the mean is unchanged, and Lemma 1.2.5 applies to the centered matrices through

$$\tilde{\mathbf{X}}_c = \mathbf{X}_c \Pi, \quad \tilde{\mathbf{X}}_c \tilde{\mathbf{X}}_c^T = \mathbf{X}_c \mathbf{X}_c^T.$$

Proceeding with the POD construction, the first m left singular vectors of $\mathbf{X}_c$ form

$$\mathbf{X}_c = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T, \quad \mathbf{U}_m = \begin{bmatrix} u_1 & \cdots & u_m \end{bmatrix}.$$

Since the matrices in this example are large, the first 200 singular triplets are computed by a randomized SVD with a fixed random seed. All smaller spaces are nested truncations of the same basis.

For a new vorticity vector ω , the centered field is $z = \omega - \bar{\omega}$. The full-field POD reference and the sparse interpolant used in the experiment are, respectively,

$$\omega_{\text{POD}} = \bar{\omega} + \mathbf{U}_m \mathbf{U}_m^T z, \tag{2.28}$$

$$\omega_{\mathbf{P}} = \bar{\omega} + \mathbf{U}_m (\mathbf{P}^T \mathbf{U}_m)^{-1} \mathbf{P}^T z. \tag{2.29}$$

Each coordinate inside $\mathbf{P}$ now has the concrete interpretation of a vorticity sensor at one mesh node. We construct $\mathbf{P}_D$ by Algorithm 1.5.1 and $\mathbf{P}_Q$ by Algorithm 1.7.1, and use (2.29) with the corresponding matrix. The linear systems are solved without explicitly forming an inverse.

Equations (2.28) and (2.29) are applied independently to the saved vorticity snapshots. No reduced differential equation is integrated after the offline construction of the mean, basis, and sensor indices.

At a fixed m , the mean $\bar{\omega}$ and basis $\mathbf{U}_m$ are identical for POD, DEIM, and QDEIM. The POD projection uses the complete field, whereas the two interpolants use only

$$\mathbf{P}_D^T \omega \quad \text{or} \quad \mathbf{P}_Q^T \omega.$$

Consequently, the comparison isolates the effect of the index-selection rule.

2.4.3 Leave-one-parameter-out experiment

The dense parameter grid is

$$\mu_i = 0.70, 0.71, \dots, 1.00.$$

To separate training and testing, the complete trajectory at $\mu = 0.85$ is withheld. The remaining 30 trajectories form

$$\mathbf{X}_{\text{train}} \in \mathbb{R}^{11994 \times 12000},$$

whereas

$$\mathbf{X}_{\text{test}} = \mathbf{W}(0.85) \in \mathbb{R}^{11994 \times 400}.$$

Neither the mean nor the POD modes contain snapshots from the test trajectory.

Figure 2.12 shows the interpolation nodes selected at $m = 150$. Both methods place many sensors near the cylinder and in the wake, where the vorticity fields exhibit their strongest spatial variation. They agree on the importance of this region while producing visibly different spatial layouts.

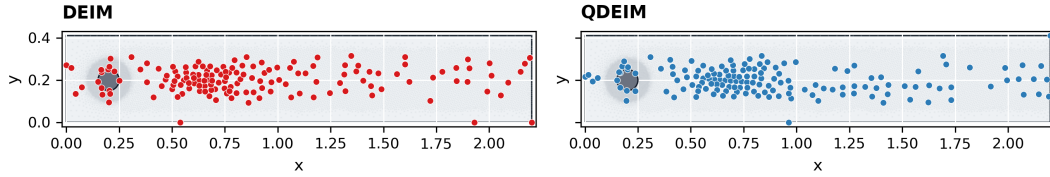


Figure 2.12: Vorticity degrees of freedom selected by DEIM and QDEIM for $m = 150$ in the leave-one-parameter-out experiment.

For a reference matrix $\mathbf{Y}$ and its reconstruction $\tilde{\mathbf{Y}}$, we report the relative Frobenius error

$$E_F = \frac{\|\mathbf{Y} - \tilde{\mathbf{Y}}\|_F}{\|\mathbf{Y}\|_F}. \quad (2.30)$$

The error is computed using the vectors of stored nodal values, so it is a discrete Euclidean error rather than a finite-element mass-matrix $L^2(\Omega)$ error.

Figure 2.13 compares the POD projection with the sparse DEIM and QDEIM reconstructions. All three errors decrease as the common dimension is increased. In this experiment QDEIM is more accurate than classical DEIM at every tested dimension, while the full-field POD projection provides the lower reference curve.

At $m = 150$, the measured errors and interpolation-matrix quantities are summarized in Table 2.2. QDEIM gives both the smaller error and the smaller interpolation constant in this test.

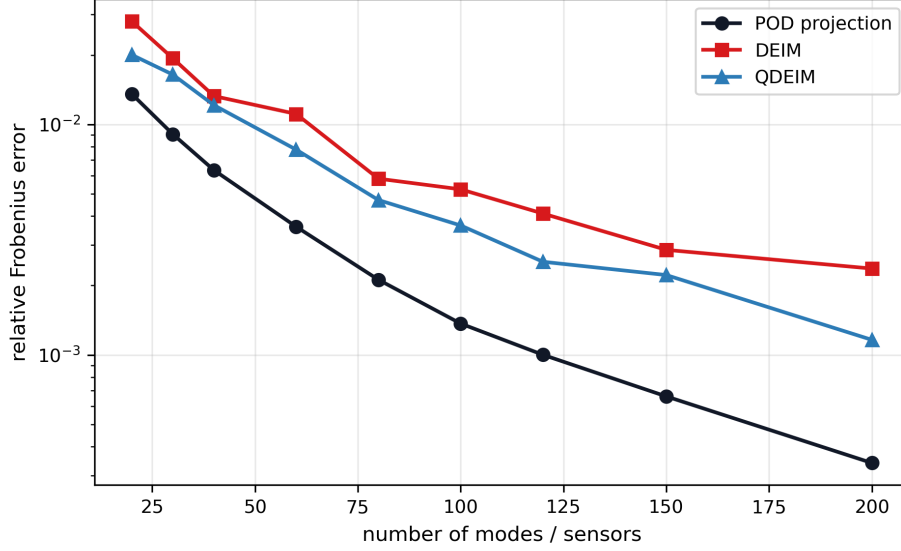


Figure 2.13: Relative Frobenius reconstruction error for the held-out trajectory $\mu = 0.85$ as the common number of POD modes and sensors is increased.

Table 2.2: Leave-one-parameter-out results at $m = 150$.

Method	E_F	$\text{cond}_2(\mathbf{P}^T \mathbf{U}_{150})$	$\ (\mathbf{P}^T \mathbf{U}_{150})^{-1}\ _2$
POD projection	$6.60 \cdot 10^{-4}$	—	—
DEIM	$2.86 \cdot 10^{-3}$	46.05	111.87
QDEIM	$2.22 \cdot 10^{-3}$	33.82	85.18

The dimension $m = 150$ is used for the external test below. It was selected from the leave-one-out experiment before the external data was evaluated. It is not claimed to be an optimal dimension; rather, it provides a representative compromise between 150 spatial measurements and an error of order 10^{-3} on the held-out grid trajectory.

2.4.4 Reconstruction at an unseen parameter

For the final test, the POD mean and basis are recomputed from all 31 parameter trajectories:

$$\mathbf{X}_{\text{train}} \in \mathbb{R}^{11994 \times 12400}.$$

The validation trajectory is generated independently at

$$\mu_{\text{ext}} = 0.855,$$

which is not a member of the training grid. The same basis and mean are used for DEIM and QDEIM. The external errors reported below use $m = 150$ sensors.

Since the validation data was generated independently, its vorticity snapshots are interpolated onto the spatial grid used to construct the POD basis before applying (2.29).

Both methods achieve an external relative Frobenius error of approximately

$$E_F \approx 2.75 \cdot 10^{-2}.$$

The difference between the two values is negligible at the accuracy relevant to this experiment, so the external test is not used to rank DEIM and QDEIM.

Figure 2.14 compares the fields at two times in the developed wake. The dimensions $m = 80, 100, 120, 150$ are shown. The alternating positive and negative vorticity structures are reproduced by both sparse interpolants throughout this range. The displayed colour scale is a common, symmetric scale chosen to reveal the wake; values close to the cylinder that exceed this range are saturated.

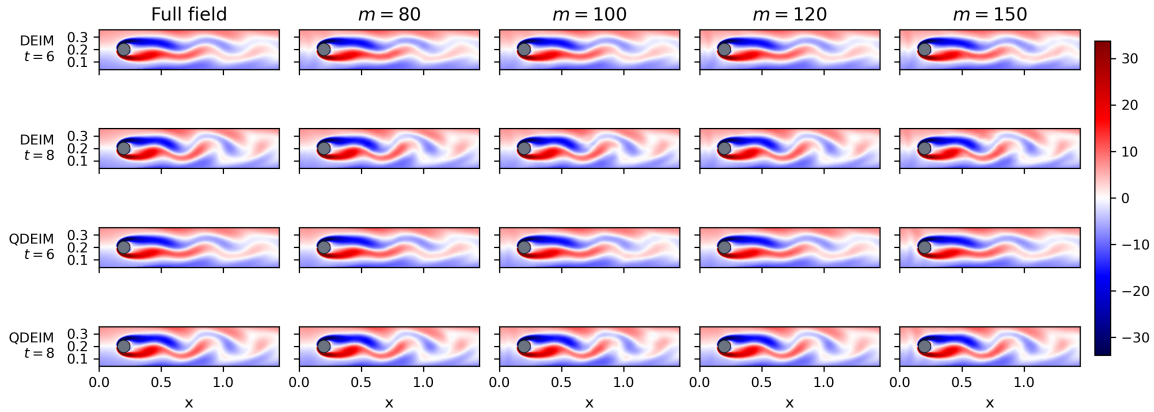


Figure 2.14: Full vorticity field and sparse DEIM and QDEIM reconstructions at $\mu = 0.855$ for $m = 80, 100, 120, 150$ at $t = 6$ and $t = 8$. A common wake-focused colour scale is used in all panels.

The external experiment shows that the spatial information learned from the training trajectories can be used to reconstruct a nearby, independently generated parameter trajectory from sparse vorticity values.

Bibliography

- [1] Maxime Barrault, Yvon Maday, Ngoc Cuong Nguyen, and Anthony T. Patera, *An “empirical interpolation” method: Application to efficient reduced-basis discretization of partial differential equations*, *Comptes Rendus. Mathématique* **339** (2004), no. 9, 667–672, <https://doi.org/10.1016/j.crma.2004.08.006>.
- [2] Peter Benner and Tobias Breiten, *Two-sided projection methods for nonlinear model order reduction*, *SIAM Journal on Scientific Computing* **37** (2015), no. 2, B239–B260, <https://doi.org/10.1137/14097255X>.
- [3] Jan Blechta, Roland Herzog, Jaroslav Hron, and Gerd Wachsmuth, *Navier–Stokes Equations*, FEniCS Hands-on Tutorial, 2018, <https://fenics-handson.readthedocs.io/en/latest/navierstokes/doc.html>, Accessed 5 August 2026.
- [4] Peter A. Businger and Gene H. Golub, *Linear least squares solutions by householder transformations*, *Numerische Mathematik* **7** (1965), no. 3, 269–276.
- [5] Saifon Chaturantabut and Danny C. Sorensen, *Nonlinear model reduction via discrete empirical interpolation*, *SIAM Journal on Scientific Computing* **32** (2010), no. 5, 2737–2764, <https://doi.org/10.1137/090766498>.
- [6] Yong Chen, *Model order reduction for nonlinear systems*, Master’s thesis, Massachusetts Institute of Technology, Cambridge, MA, 1999, <https://hdl.handle.net/1721.1/9381>.
- [7] Youngsoo Choi, Deshawn Coombs, and Robert Anderson, *SNS: A solution-based nonlinear subspace method for time-dependent model order reduction*, *SIAM Journal on Scientific Computing* **42** (2020), no. 2, A1116–A1146, <https://doi.org/10.1137/19M1242963>.
- [8] Emily Clark, J. Nathan Kutz, and Steven L. Brunton, *Sensor selection with cost constraints for dynamically relevant bases*, *IEEE Sensors Journal* **20** (2020), no. 19, 11674–11687, <https://doi.org/10.1109/JSEN.2020.2997298>.

- [9] Zlatko Drmač and Serkan Gugercin, *A new selection operator for the discrete empirical interpolation method—improved a priori error bound and extensions*, SIAM Journal on Scientific Computing **38** (2016), no. 2, A631–A648, <https://doi.org/10.1137/15M1019271>.
- [10] Zlatko Drmač, Vjeran Hari, Miljenko Marušić, Mladen Rogina, Sanja Singer, and Saša Singer, *Numerička analiza: Predavanja i vježbe*, Sveučilište u Zagrebu, Prirodoslovno-matematički fakultet, Matematički odjel, Zagreb, 2003, <https://web.math.pmf.unizg.hr/nastava/unm/materijali/Hari%20--%20Numericka%20analiza%20-%20osnovni%20udzbenik.pdf>, Accessed 12 August 2026.
- [11] Benjamin Peherstorfer, Zlatko Drmač, and Serkan Gugercin, *Stability of discrete empirical interpolation and gappy proper orthogonal decomposition with randomized and deterministic sampling points*, SIAM Journal on Scientific Computing **42** (2020), no. 5, A2837–A2864, <https://doi.org/10.1137/19M1307391>.
- [12] The MORwiki Community, *Nonlinear RC ladder*, MORwiki—Model Reduction Wiki, 2018, https://modelreduction.org/morwiki/Nonlinear_RC_Ladder.

Sažetak

Svrha ovog diplomskog rada bila je prikazati diskretnu empirijsku interpolacijsku metodu (DEIM) i proučiti dvije odluke na kojima se ona temelji: odabir aproksimacijskog potprostora i odabir interpolacijskih indeksa. Prava ortogonalna dekompozicija (POD) omogućuje konstrukciju modela smanjenog reda projekcijom na niskodimenzionalni potprostor. Ipak, u prisutnosti opće nelinearnosti izračun desne strane i dalje zahtijeva povratak na punu dimenziju sustava. DEIM rješava taj preostali problem aproksimacijom nelinearne vektorske funkcije u niskodimenzionalnom potprostoru, pri čemu se koeficijenti aproksimacije određuju iz malog broja odabranih koordinata.

U teorijskom dijelu rada razvijeni su alati linearne algebre potrebni za konstrukciju svih obrađenih metoda. POD se temelji na dekompoziciji singularnih vrijednosti (SVD), dok interpolacijska matrica u DEIM-u određuje kako se računaju koeficijenti aproksimacije. Izvedena je ocjena pogreške DEIM aproksimacije koja razdvaja utjecaj tih dviju odluka: jedan faktor mjeri dio vektora koji nije obuhvaćen odabranim potprostorom, a drugi pojačanje pogreške uzrokovano odabirom interpolacijskih indeksa. Obrađena je i inačica DEIM-a pod nazivom QDEIM, koja mijenja pravilo odabira indeksa te primjenjuje QR faktorizaciju s pivotiranjem stupaca. QDEIM koristi isti aproksimacijski potprostor kao DEIM, ali općenito odabire druge indekse i određuje interpolacijsku matricu drugačije uvjetovanosti.

Uz DEIM i QDEIM, u radu su obrađene još dvije modifikacije postupka. Prva ograničava dopuštene indekse kada se mjerenja ili evaluacije funkcije ne mogu provoditi na dijelu domene. Dopuštena interpolacijska matrica postoji točno kada retci baze koji odgovaraju dopuštenim indeksima čine matricu punog stupčanog ranga. Taj uvjet jamči postojanje, ali ne i dobru uvjetovanost. Druga je metoda SNS, koja mijenja konstrukciju nelinearnog aproksimacijskog potprostora u *offline* fazi. Umjesto prikupljanja uzoraka nelinearne funkcije, taj se potprostor konstruira iz proširene baze prostora stanja. Budući da ograničeni odabir i SNS djeluju na različite dijelove postupka, mogu se međusobno kombinirati.

Numerički dio započinje usporedbom DEIM-a i QDEIM-a na parametriziranoj familiji funkcija. Dobivene su usporedive rekonstrukcije, iako nisu odabrani isti interpolacijski indeksi. Zatim je na primjeru s nedopuštenim dijelom domene ograničenje skupa indeksa povećalo interpolacijsku konstantu i pogrešku rekonstrukcije. Rezultati potvrđuju da uvjet ranga i uvjetovanost pripadne podmatrice opisuju različita svojstva ograničenog interpolacijskog problema.

Primjer FitzHugh-Nagumova sustava pokazao je da mali reziduali POD aproksimacija stanja i nelinearne funkcije sami po sebi ne jamče jednoliku malu pogrešku vremenski integrirane reducirane trajektorije. Mali vremenski pomak brzog prijelaza može uzrokovati veliku trenutačnu pogrešku stanja, čak i kada promatrane komponente rješenja ostaju bliske. Reduciranu diferencijalnu jednadžbu stoga treba ocjenjivati prema njezinoj simuliranoj trajektoriji, a ne samo prema singularnim vrijednostima matrica uzoraka.

Primjer nelinearnog RC-ladder sklopa pokazao je još jednu praktičnu činjenicu. Učinkovita evaluacija DEIM aproksimacije ne zahtijeva da svaka komponenta nelinearnog vektora ovisi samo o pripadnoj komponenti stanja. Dovoljno je da se odabrane komponente mogu izračunati iz malog lokalnog skupa potrebnih koordinata. U ovom je primjeru taj skup određen točno pa je uzorkovano vektorsko polje izračunano bez formiranja cijelog nelinearnog vektora. Standardne metode i inačice zasnovane na SNS-u konvergirale su s povećanjem dimenzije, dok je njihov relativni poredak ovisio o odabranoj mjeri pogreške. Eksperiment ne izdvaja jednu inačicu kao općenito najbolju, ali pokazuje da SNS može izbjeći SVD matrice uzoraka nelinearne funkcije i pritom omogućiti preciznu hiperredukciju.

Konačno, u primjeru vrtložnog toka iza cilindra DEIM i QDEIM korišteni su za rekonstrukciju iz rijetkih mjerenja, a ne kao dio vremenski integriranog reduciranog modela. POD baza i položaji senzora određeni su u *offline* fazi, nakon čega su polja vrtložnosti za nove vrijednosti parametra rekonstruirana iz malog broja mjerenja. Obje su metode reproducirale glavne strukture vrtložnog traga na neovisno generiranoj trajektoriji. Taj primjer povezuje interpolacijske indekse s fizičkim položajima senzora te naglašava važnost jasnog navođenja onoga što se u pojedinom primjeru reducira ili rekonstruira.

Svi rezultati pokazuju da pravilo odabira interpolacijskih indeksa nije jedini čimbenik uspješnosti metode tipa DEIM. Ona ovisi o kvaliteti aproksimacijskog potprostora, stabilnosti i dopustivosti odabranih indeksa, trošku izračuna potrebnih komponenti te, kod dinamičkih sustava, načinu na koji se učinak aproksimacije širi kroz vrijeme. Odvojeno promatranje tih uloga daje koristan okvir za odabir i tumačenje DEIM-a, QDEIM-a, ograničenog odabira indeksa i SNS-a u konkretnoj primjeni.

Summary

The purpose of this thesis was to present the discrete empirical interpolation method and examine the two decisions on which it is based: the construction of an approximation space and the selection of interpolation indices. A POD projection can substantially reduce the dimension of a state, but it does not generally reduce the cost of evaluating a nonlinear vector in the original space. DEIM addresses this remaining cost by approximating that vector in a low-dimensional basis and determining its coefficients from only a small number of selected entries.

The theoretical part developed the linear-algebraic tools needed for this construction. The SVD provides the POD approximation spaces, while the interpolation matrix determines how the basis coefficients are recovered. The DEIM error estimate separates these two effects: one factor measures the part of the vector not represented by the chosen basis, and the other measures the amplification introduced by the selected indices. QDEIM changes the index-selection rule by applying column-pivoted QR to the transposed basis. It therefore uses the same approximation space as DEIM but may select different coordinates and produce an interpolation matrix with different stability.

The same viewpoint also clarifies the two further modifications considered in the thesis. When measurements or evaluations are possible only on a prescribed part of the domain, the index search can be restricted to an admissible set. An admissible interpolation matrix exists precisely when the corresponding rows of the basis retain full column rank. This rank condition guarantees existence, but not good conditioning. SNS modifies a different part of the offline procedure. Instead of constructing the nonlinear basis from a separate matrix of nonlinear snapshots, it uses an extended solution basis and the time-discrete equations. The basis-construction rule and the index-selection rule are therefore independent choices and can be combined in different ways.

The numerical examples illustrated these distinctions. For the parameter-dependent functions, DEIM and QDEIM produced comparable reconstructions even though their interpolation points were not identical. Restricting the admissible points preserved solvability in the tested cases, but increased the interpolation constant and the reconstruction error. This confirmed that the rank test and the conditioning of the selected submatrix describe different properties of the restricted problem.

The FitzHugh–Nagumo example showed that small POD and nonlinear-snapshot residuals do not by themselves guarantee a uniformly small error in an integrated reduced trajectory. A small displacement in the time at which a rapid transition occurs can cause a pronounced instantaneous state error even when the observed solution traces remain close. The reduced differential equation must therefore be assessed through its simulated trajectory, not only through the singular values of the training snapshot matrices.

The nonlinear RC ladder demonstrated another practical point. Efficient DEIM evaluation does not require every component of the nonlinear vector to depend only on the corresponding state component. It is sufficient that the selected entries can be evaluated from a small dependency closure. In this example the required local stencil was determined exactly, so the sampled vector field was computed without forming the complete nonlinear vector. The conventional and SNS-based variants all converged as their dimensions increased, while their relative ordering depended on the chosen error measure. The experiment does not identify a universally preferable variant, but it shows that SNS can avoid the nonlinear-snapshot SVD without preventing accurate hyper-reduction.

Finally, the cylinder-wake example used DEIM and QDEIM as sparse reconstruction methods rather than as parts of a time-integrated reduced model. The POD basis and sensor locations were fixed offline, after which vorticity fields at unseen parameter values were reconstructed from sparse measurements. Both methods reproduced the principal wake structures on an independently generated trajectory. This experiment connects interpolation indices with physical sensor locations and also emphasizes the importance of stating clearly which reduced or reconstructed quantity is being tested.

Taken together, the results show that the performance of a DEIM-type method cannot be attributed to the interpolation rule alone. It depends on the quality of the approximation space, the stability and feasibility of the selected indices, the cost of evaluating the required entries, and, for dynamical systems, the propagation of the resulting approximation through time. Keeping these roles separate provides a useful framework for choosing and interpreting DEIM, QDEIM, restricted selection, and SNS in a particular application.

Curriculum Vitae

I was born on 20 September 2002 in Rijeka, where I attended Nikola Tesla Elementary School and later completed the natural sciences and mathematics programme at Prva sušačka hrvatska gimnazija. I graduated in 2021 as a nominee for the school's Best Student Award.

I then moved to Zagreb to study mathematics at the Faculty of Science, University of Zagreb, where I completed my bachelor's degree in 2024. Having developed a particular interest in the interplay between pure mathematics and numerical analysis, I decided to continue my studies by enrolling in the master's programme in Applied Mathematics at the same faculty, which I am now completing with this thesis.

During my master's studies, I received the award for the best student across all mathematics master's programmes at the Faculty of Science and participated in the Guest Student Programme at the Jülich Supercomputing Centre in Germany from August to October 2025. During the programme, I worked on the project *Implementing Task Parallelism with HPX and Kokkos for Lattice QCD*, under the supervision of Dr Travis Whyte. The project focused on exploring asynchronous task-based parallelism using the HPX runtime system together with Kokkos as a performance-portable backend, with the broader goal of investigating their combined use in high-performance scientific computing.